TDR-300B PLUS

Microorganism Analysis System

Operator’s Manual

A Mindray Company

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- The product is used in accordance with the instructions for use.

WARNING:

It is important for the hospital or organization that employs this equipment to carry out a reasonable service/maintenance plan. Neglect of this may result in machine damage or personal injury.

NOTE:

This equipment must be operated by skilled/trained clinical professionals.

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Customer Service Department

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Preface

Thank you for purchasing the Microorganism Analysis System.

Before using the microorganism analysis system, please read this manual thoroughly and understand it for relevant operation instructions.

Please keep this manual properly for convenient use.

Who Should Read This Manual

This manual is written for clinical laboratory professionals to:

Perform daily operating tasks;

Perform system maintenance and troubleshooting;

Learn about the system hardware and software.

WARNING:

The device is to be operated only by clinical professionals, doctors or experimenters trained by our company or our authorized distributors.

What Can You Find in This Manual

This manual covers principles, operations, daily maintenance and troubleshooting of the system. Please operate and service the system strictly as instructed by this manual.

Conventions Used in This Manual

This manual uses certain typographical conventions to clarify meanings in the text.

Bold font indicates a chapter title, such as **5 Trouble shooting.**

Safety Symbols, Labels and Silkscreen

Safety symbols, labels and silkscreen are used in this manual in order to remind you of the potential hazards and risks in operating this product. The symbols are used in combination with relevant texts.

1. Safety symbols	
	WARNING Read the statement following the symbol. The statement is alerting you to an operating hazard that can cause personal injury.
	CAUTION Read the statement following the symbol. The statement is alerting you to a possibility of system damage or unreliable results.
	Note: Read the statement following the symbol. The statement is alerting you to information that requires your attention.

	BIOHAZARD Read the statement following the symbol. The statement is alerting you to a potentially biohazardous condition.
2. Labels and Silkscreen	
SN	Serial number
	Date of manufacture
	Manufacturer
IVD	In Vitro diagnostic equipment
20	Environment-friendly use period (20 years)
CE	This product is provided with a CE marking in accordance with the regulations stated in Directive 98/79/EC concerning medical devices.
	BIOHAZARD Read the statement following the symbol. The statement is alerting you to a potentially biohazardous condition.
WARNING Risk of Electrical Shock	Electric Shock Alerting the user of electric shock. Note: Do not remove the shell of the product; otherwise you may get electrically shocked.
WARNING MOVING PARTS Do not touch when in operation.	Note: Personal injury or equipment damage may be caused.
	Electrical ground
|	On (Main power)
○	Off (Main power)
IOIOI	Serial Port
	USB Interface

	Storage Temperature
	Storage Humidity
	Storage Atmospheric Pressure

Summary of Hazards

To use the product safely and efficiently, pay attention to the following safety precautions. Ignoring any of these safety precautions may lead to personal injury or equipment damage.

WARNING

If the product is used in a manner not specified by our company, the protection provided by the product may be invalid.

If you want to relocate your system, contact our customer service department or your local distributor.

Electric Shock Hazards

Observe the following instructions to prevent electric shock.

WARNING

When the MAINS POWER is turned on, users other than the servicing personnel authorized by our company must not open the rear cover.

Spillage of solutions on the product may cause equipment failure and even electric shock. Do not place specimen and reagent on the product. In case of spillage, switch off the power immediately, remove the spillage and contact our customer service department or your local distributor.

Moving Parts Hazards

Observe the following instructions to prevent personal injury caused by moving parts.

WARNING

When the system is in operation, do not touch such moving parts

Do not put your finger or hand into any open part when the system is in operation.

Fire and Explosion Hazards

Observe the following instructions to prevent fire and explosion.

WARNING

Ethanol is flammable substance. Please exercise caution while using ethanol.

Electromagnetic Noise Precautions

CAUTION

Electromagnetic noise may interfere with operations of the system. Do not install devices generating excessive electromagnetic noise around the system. Do not use such devices as radio transmitters in the room housing the system. Do not use other CRT displays around the system.

Do not use other medical instruments around the system that may generate electromagnetic noise to interfere with their operations.

Do not use this device in close proximity to sources of strong electromagnetic radiation (e.g. mobile phones or radio transmitters), as these may interfere with the proper operation.

The electromagnetic environment should be evaluated prior to operation of the device.

This device has been designed and tested to CISPR 11 Class A, and in a domestic environment may cause radio interference, in which case, you may need to take measures to mitigate the interference.

It is the manufacturer's responsibility to provide equipment electromagnetic compatibility information to the customer or user.

It is the user's responsibility to ensure that a compatible electromagnetic environment for the equipment can be maintained in order that it will perform as intended.

Biohazards

Observe the following instructions to protect against the biohazardous infection by samples, calibrators and control samples.

BIOHAZARD

Inappropriate handling of specimens may lead to biohazardous infection. Do not touch specimens directly with your hands. Wear gloves and lab coat, if necessary, goggles.

In case your skin contacts the specimens and mixtures, follow standard laboratory safety procedure and consult a doctor.

Waste Disposal

Observe the following instructions to prevent environmental pollution and personal injury caused by waste.

BIOHAZARD

Dispose of waste parts or the whole unit in accordance with your local or national rule for waste disposal. Wear gloves and lab coat while handling them.

NOTE:

The definition of the WEEE label applies to EU member states only: The use of this symbol indicates that this system should not be treated as household waste. By ensuring that this system is disposed of correctly, you will help prevent bringing potential negative consequences to the environment and human health. For more detailed information with regard to returning and recycling this system, please consult the distributor from whom you purchased the system.

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1 System Description

1.1 Intended Scope

This device is intended for microorganism identification and susceptibility test in medical and health service, epidemic prevention institute and research institute. The bacteria that can be identified by the device include: Enterobacteriaceae, Non-fermentative Bacteria, Micrococcaceae, Streptococcus, Yeast-Like Fungi, Vibrionaceae, Coryneform bacteria, Neisseria & Haemophilus, Bacillus, and Anaerobe.

1.2 Performance and Technical Specifications

Specifications	Parameter
Bacteria type:	Enterobacteriaceae, Non-fermentative Bacteria, Micrococcaceae, Streptococcus, Yeast-Like Fungi, Vibrionaceae, Coryneform bacteria, Neisseria & Haemophilus, Bacillus, and Anaerobe.
Test principle	Colorimetry and turbidity
Test speed:	60-80 cards/hour
Test wavelength:	R(red), G(green) and B(blue)
Test panel	96 holes
Language:	English
Recording method:	Liquid crystal display (LCD) and external printer
Display monitor:	LCD display
Print:	Printer
Communication Interface:	RS-232
Operating temperature and relative humidity:	Operating temperature: 10℃-30℃ Relative humidity: 35%-80%, without condensation
Atmospheric pressure:	76kPa - 106kPa
Transportation and storage condition:	Temperature: -40°C~40°C Relative humidity: 30%-80%, without condensation Atmospheric pressure: 76kPa - 106kPa

Specifications	Parameter
Operating environment:	The device should be placed on a steady and flat table, away from direct strong sun light and electromagnetic interference, and well grounded.
Power supply:	100~240V, 50/60Hz three-wire power cord and properly grounded
Fuse:	Φ5mm×20mm，250V 2A GLS
Input power:	no greater than 100VA
Safety classification:	Electric shock prevention: I; Overvoltage category: II; pollution degree:2
Dimensions and weight:	680mm(L)×450mm(W)×780mm(H) N.W: 21kg, G.W.: 26kg

1.3 EMC Requirements

Do not use this device in close proximity to sources of strong electromagnetic radiation (e.g. unshielded intentional RF sources), as these may interfere with the proper operation.

This equipment complies with the emission and immunity requirements of the EN61326-1:2006 and EN61326-2-6:2006.

This equipment has been designed and tested to CISPR11 Class A. In a domestic environment it may cause radio interference, in which case, you may need to take measures to mitigate the interference.

Note1:

It is the manufacturer's responsibility to provide equipment electromagnetic compatibility information to the customer or user.

Note2:

It is the user's responsibility to ensure that a compatible electromagnetic environment for the equipment can be maintained in order that the device will perform as intended.

2 Installation Requirements and Procedure

2.1 Accessory

The device is packed paper skin. Please remove the package with caution. Check the package and device, which are supposed to be intact. Check the packing list. If you find any damage or omission, please contact the agent or the distributor.

The accessory list of the device is shown in Table 2.1.

Content	Quantity
Packing List	2
Operator’s Manual	1
Operation Note	1
Software dog	1
USB to Serial port Cable	1
Power cord	1
Fuse	2
USB flash drive	1
Barcode scanner	1

Table 2.1

2.2 Operating Environment

The Microorganism Analysis System should be installed in the following laboratory environment:

Ambient temperature: 10℃-30℃.

Humidity: 35%RH~80%RH.

Atmospheric pressure: 76 kPa - 106kPa.

Power supply: Voltage: AC 100～240V；Power supply frequency: 50/60 Hz；Good protective grounding.

Limit of the table：≥100kg.

Away from dust, direct sun light, splashed liquid, erosive gas, vibration, and strong electromagnetic interference.

In addition, the Microorganism Analysis System should be separated from the Microorganism analysis laboratory and not placed together with other electric devices in order to prevent interference.

WARNING

To ensure personal safety and normal operation of the device, the earth wire of the power socket must be well grounded. Using a two-wire plug board is forbidden.

The device is for intended use and must not be opened without permission. The power switch, main control board, connecting wire, and test table can be checked and supplied only by the manufacturer.

In addition, the space for holding the device must be larger than its size. For TDR-300B Plus, the clearance requirement is shown in Figure 2.1.

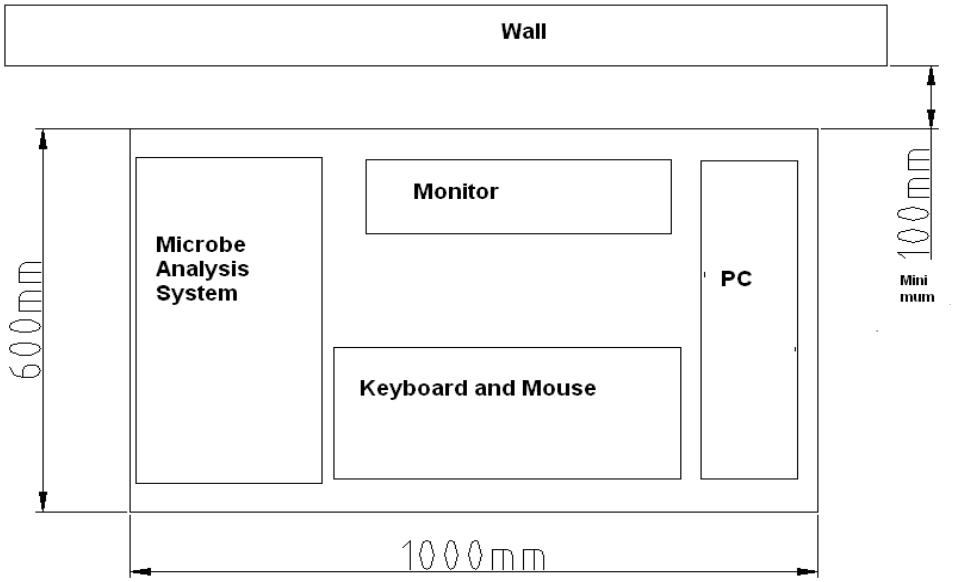


Figure 2.1 Clearance Requirement of Microorganism Analysis System

2.3 Installation Requirements and Procedure

Computer installation: Lock the display monitor into the base and place it on the table. Follow the operation manual to connect the signal line to the computer.

Soft dog installation: Insert the soft dog to the USB interface of the computer.

Device installation: Connect the USB to serial port cable between the computer and the device, and then connect the power cord.

Power supply check:

Before powering on, check if the lines are properly connected. After confirming the connection is correct, power on the device. If the device can reset automatically and the indicator light is green, it indicates that the device has been installed correctly.

Please refer to the file in USB flash drive to get the detailed installation procedure :

H-7905-CTO-03 TDR300B PLUS（International）Installation Guide

Note：

The system should be installed only by technicians of or authorized by our company.

2.4 Test Principle

Test principle: The light from the lamp irradiates on the test card and penetrates it to enter the detector. After processing of the circuit, the device tests and analyzes the biochemical holes and susceptibility test holes on the test card.

2.5 Startup the System

Connect the power cord and toggle the switch to the "｜" position. The system starts initialization and self-check, and the indicator is yellow. When the system completes the self-check, it enters standby status and the indicator turns to green.

Note:

If the indicator is not green after self-check is finished, please contact the service department for repairing.

2.6 Power Off

Toggle the power switch to the "○" position to power off the system.

Before powering off the system, make sure the test has been finished and test card taken out.

3 Software Operation

3.1 Login

Turn on the computer. When the operating system is started up, double click the Microorganism Analysis System shortcut icon. The login window pops up as shown in the following figure.

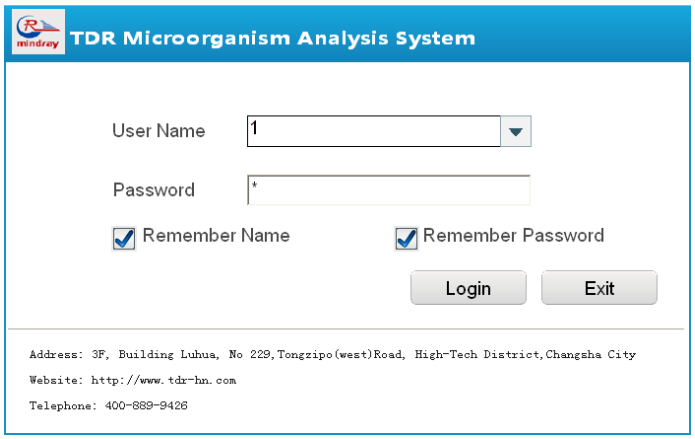


Fig 3-1

Select a ***username*** from the drop-down list and enter ***password*** to enter the main interface.

Note: Username and password for first login:

Default username: 1

Default password: 1

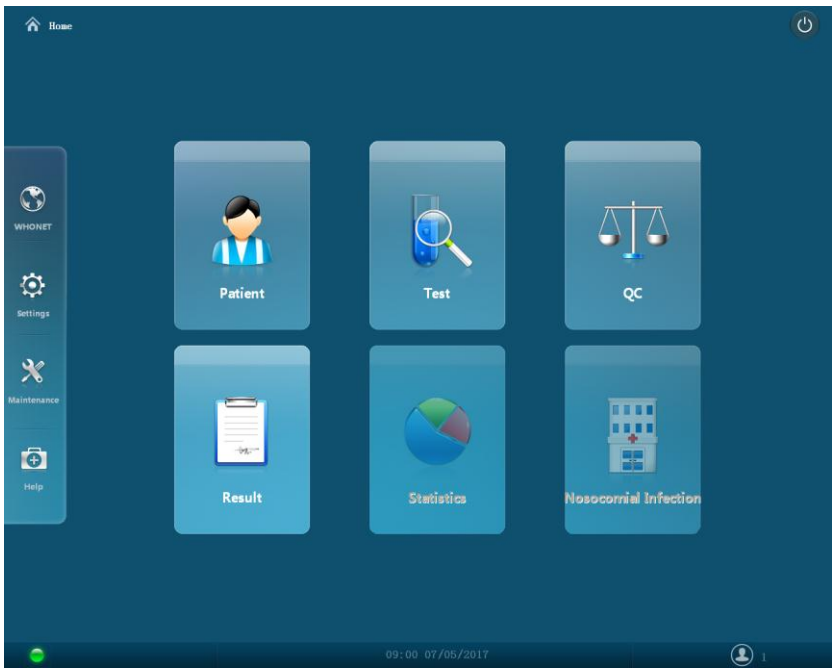


Fig 3-2

3.2 Setting

This function provides settings of ***basic setting***, ***user parameters***, ***card type***, etc.
Click the “Settings” button on the main interface; it will go to the settings module as below:

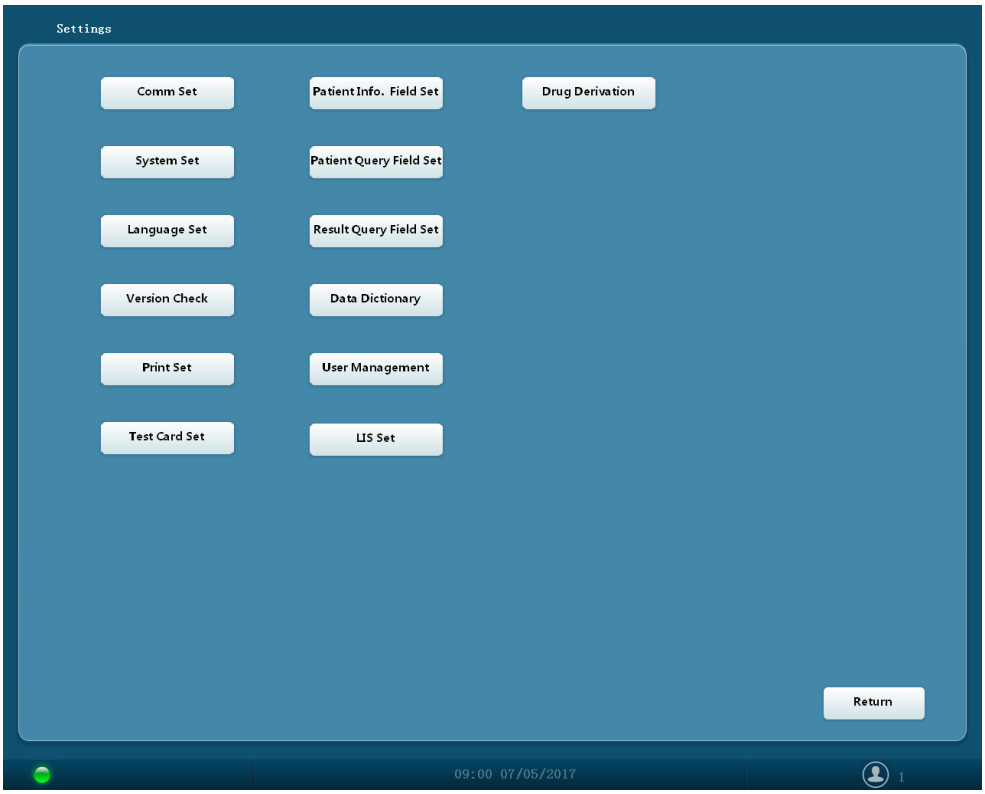
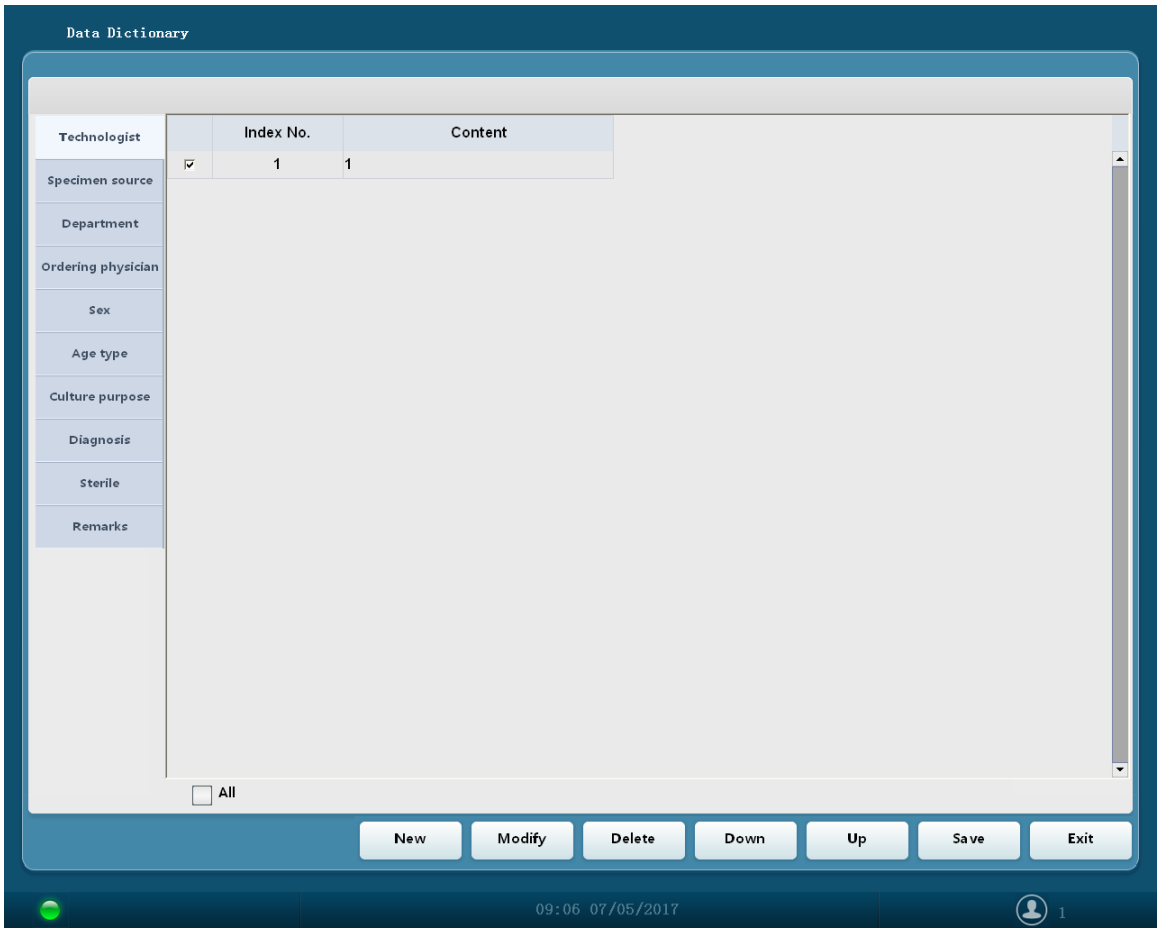


Fig 3-3

3.2.1 Basic Setting

Click "Data Dictionary" to enter the basic information interface;

3.2.1.1 Technologist Setting



Operations：

- Menu

are function keys;

- Click “New” button to enter the input window, the index No. will be generated automatically, input the technologist's name.

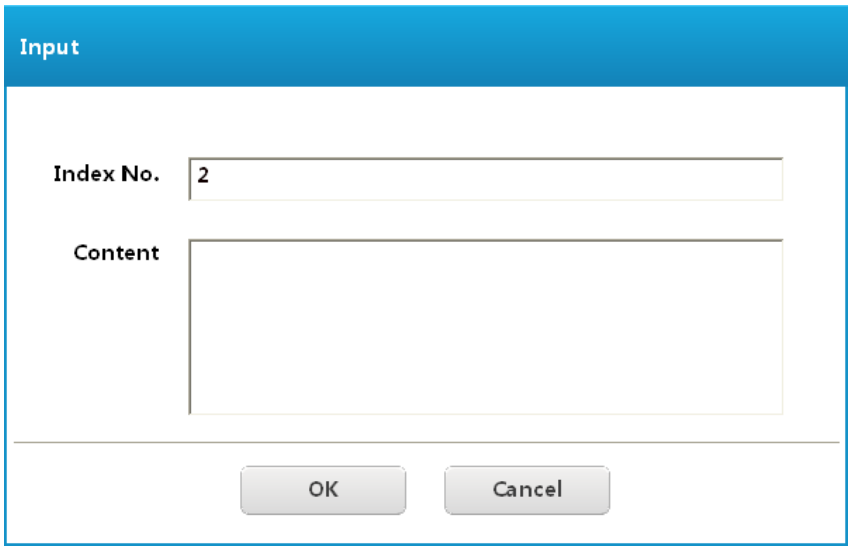


Fig 3–4

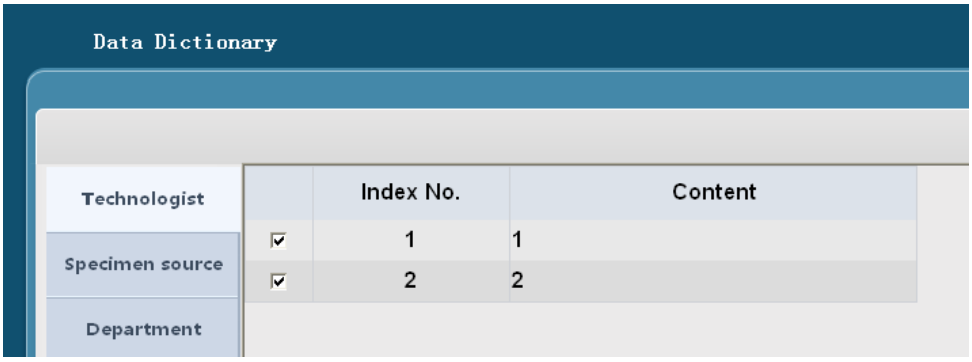


Fig 3–5

- Click “OK” button , the content is added successfully;
- If you want to enable the option, check "√" before the option to enable it;
- If you want to set the technologist's permission, return to settings interface; click "User Management" to set the permission and password for technologist.

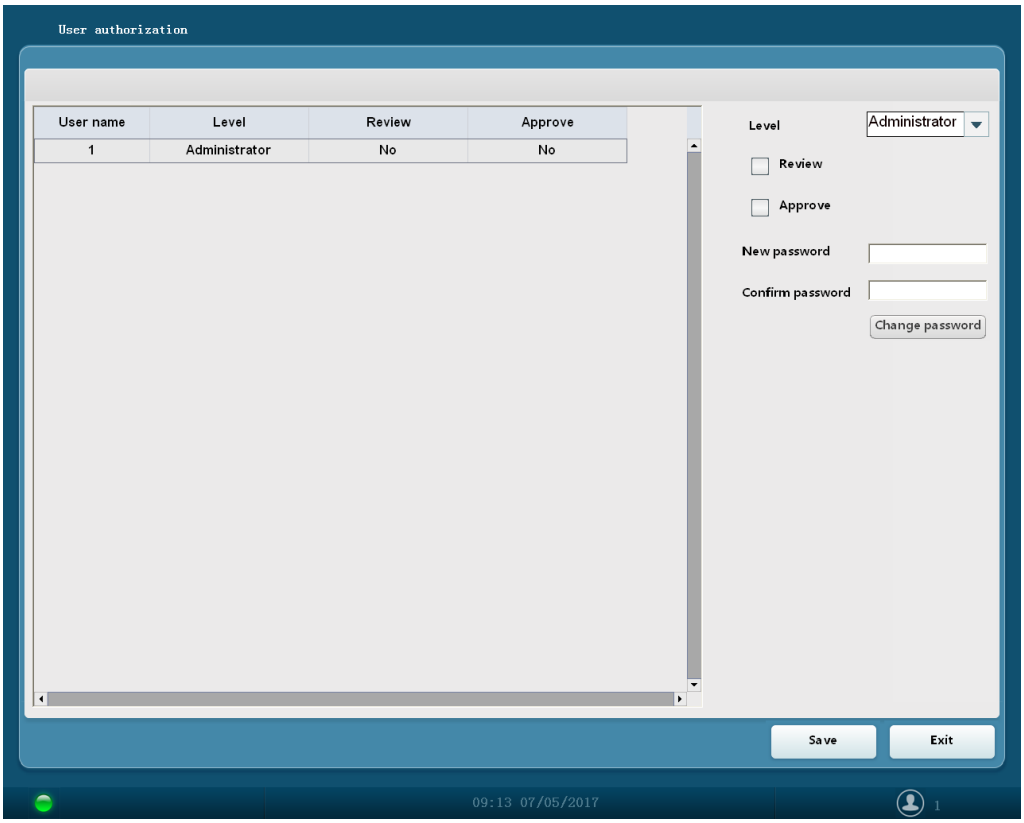


Fig 3–6

3.2.1.2 Specimen source Setting：

Click “Data Dictionary->Specimen source”, the specimen source setting interface is showed as below:

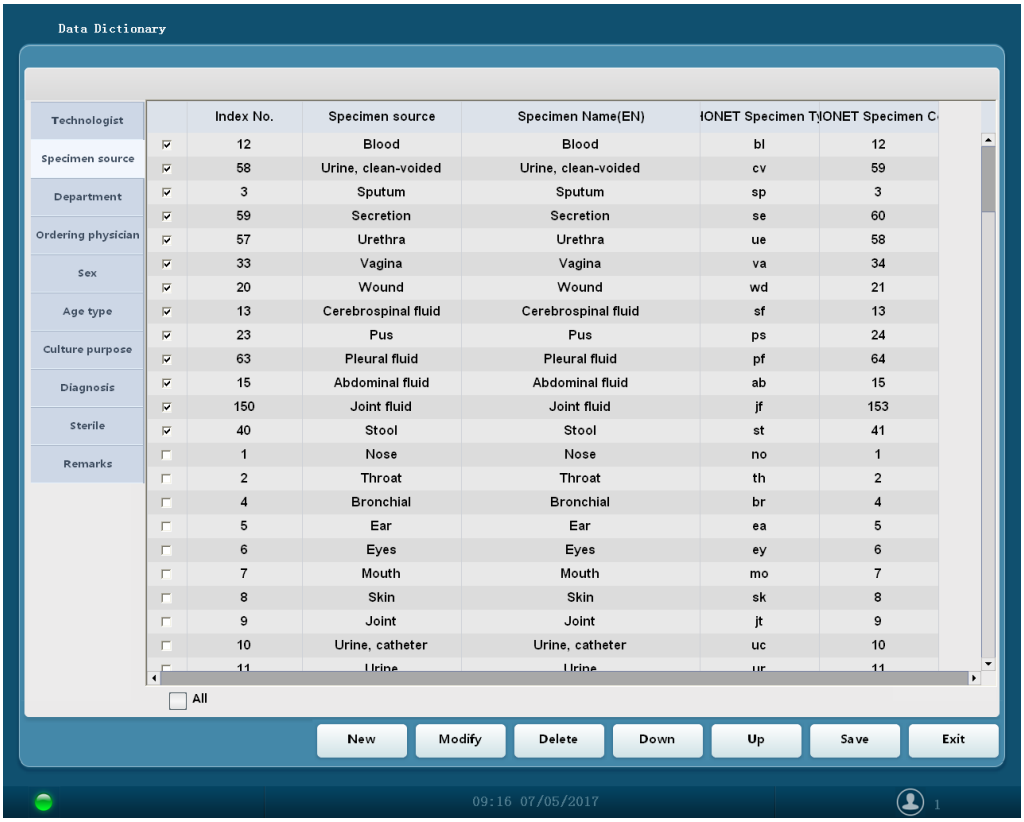
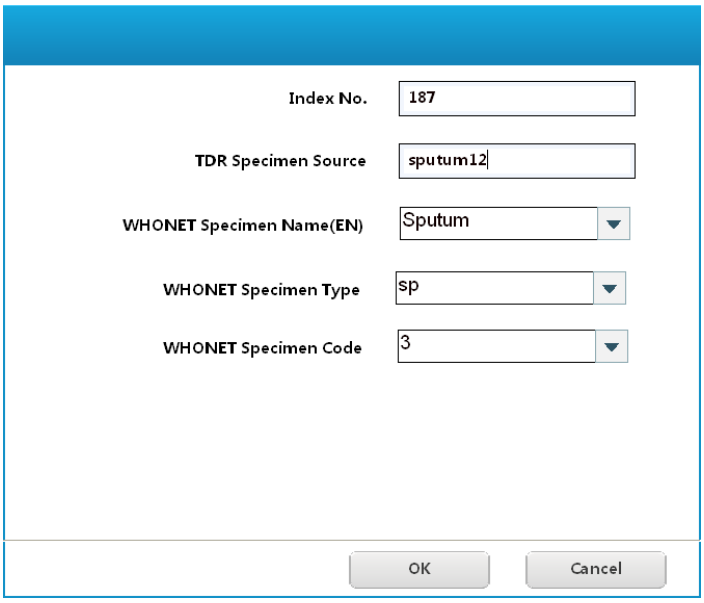


Fig 3–7

- The existing specimen sources are the largest collection of specimen sources, exactly matching the code in the WHONET database.
- User can check the "√" to enable the specimen source option;
- User can click the "Down" or "Up" button to sort the specimen source list, click "Modify" button to edit the specimen content.
- User can click the "New" button to add new specimen, the index No. will be generated automatically, input the specimen name in the TDR specimen source input line, select the corresponding WHONET Specimen Name (EN) in the drop-down list, or input the part or complete name in the input line (System will search the matched name, then you can choose the target one), then click "OK".

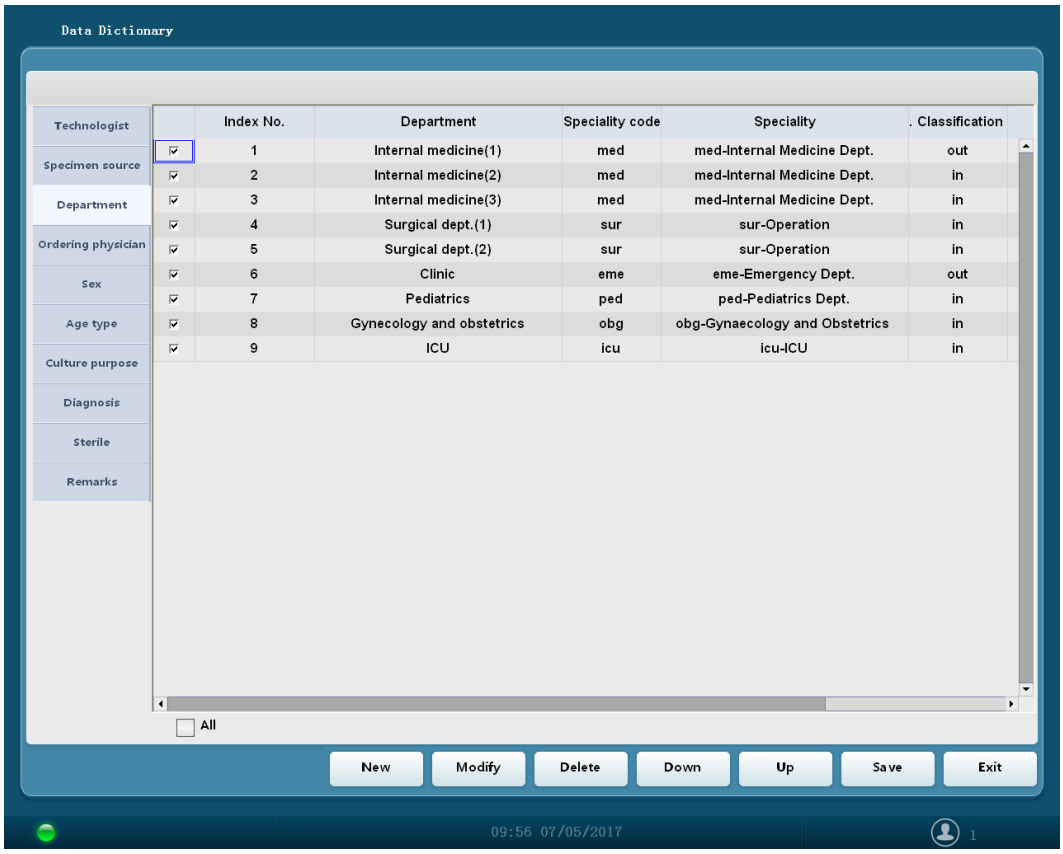


3.2.1.3 Department Setting

Click “Data Dictionary->Department”, the department setting interface is showed as below:

Operations:

- The department setting interface has listed all department names and the matched code in WHONET database.
- Click "New" button to add new department, the index No. will be generated automatically, input the department name, and select the corresponding WHONET speciality and department classification.



	Index No.	Department	Speciality code	Speciality	. Classification
☑	1	Internal medicine(1)	med	med-Internal Medicine Dept.	out
☑	2	Internal medicine(2)	med	med-Internal Medicine Dept.	in
☑	3	Internal medicine(3)	med	med-Internal Medicine Dept.	in
☑	4	Surgical dept.(1)	sur	sur-Operation	in
☑	5	Surgical dept.(2)	sur	sur-Operation	in
☑	6	Clinic	eme	eme-Emergency Dept.	out
☑	7	Pediatrics	ped	ped-Pediatrics Dept.	in
☑	8	Gynecology and obstetrics	obg	obg-Gynaecology and Obstetrics	in
☑	9	ICU	icu	icu-ICU	in

Fig 3–8

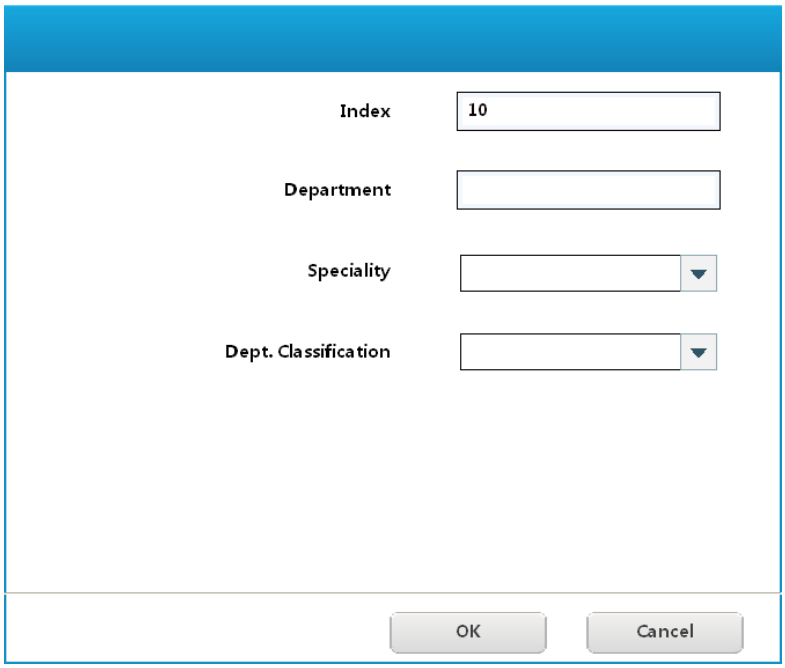


Fig 3–9

3.2.1.4 Ordering Physician Setting

Operations: Users can add, modify and delete ordering physicians;

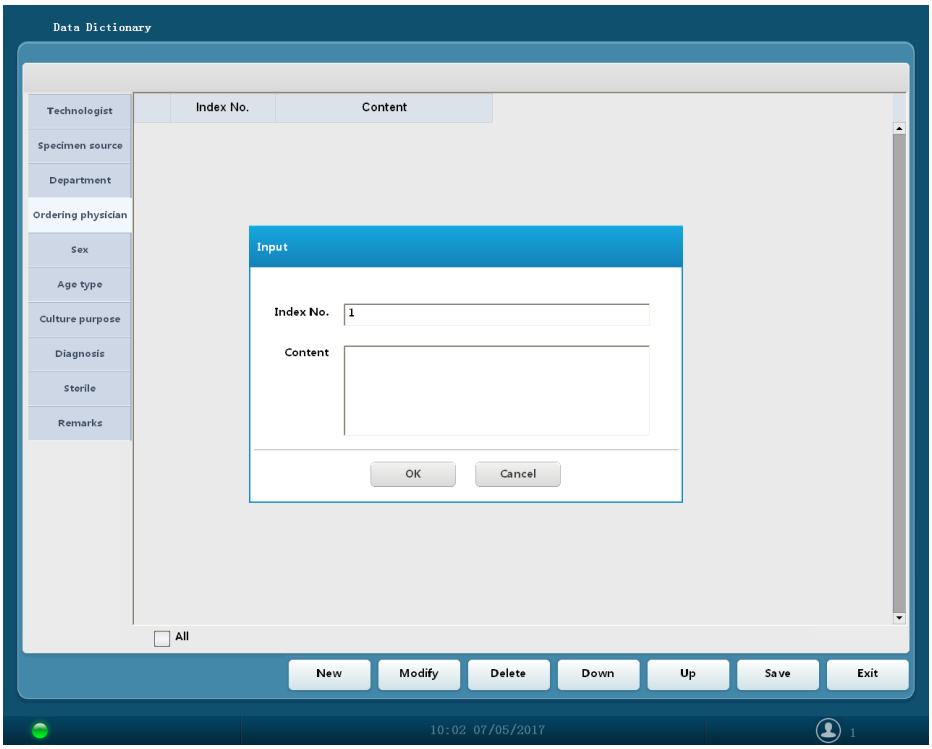


Fig 3-10

3.2.1.5 Sex Setting

Sex setting：System default settings;

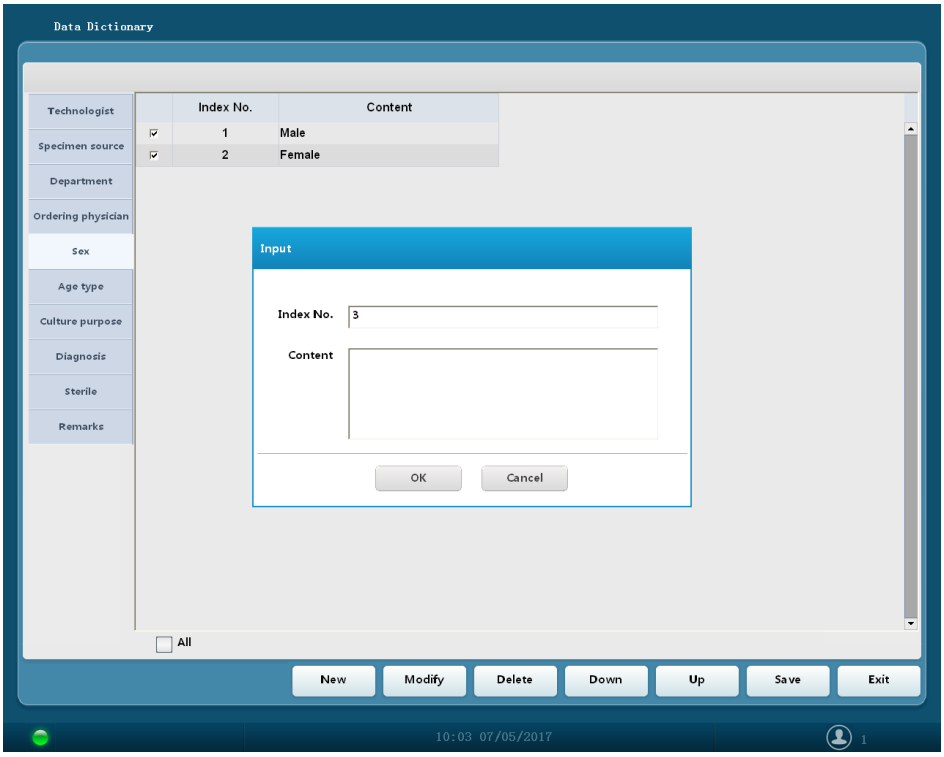


Fig 3-11

3.2.1.6 Age Type Setting

Age type setting：System default setting;

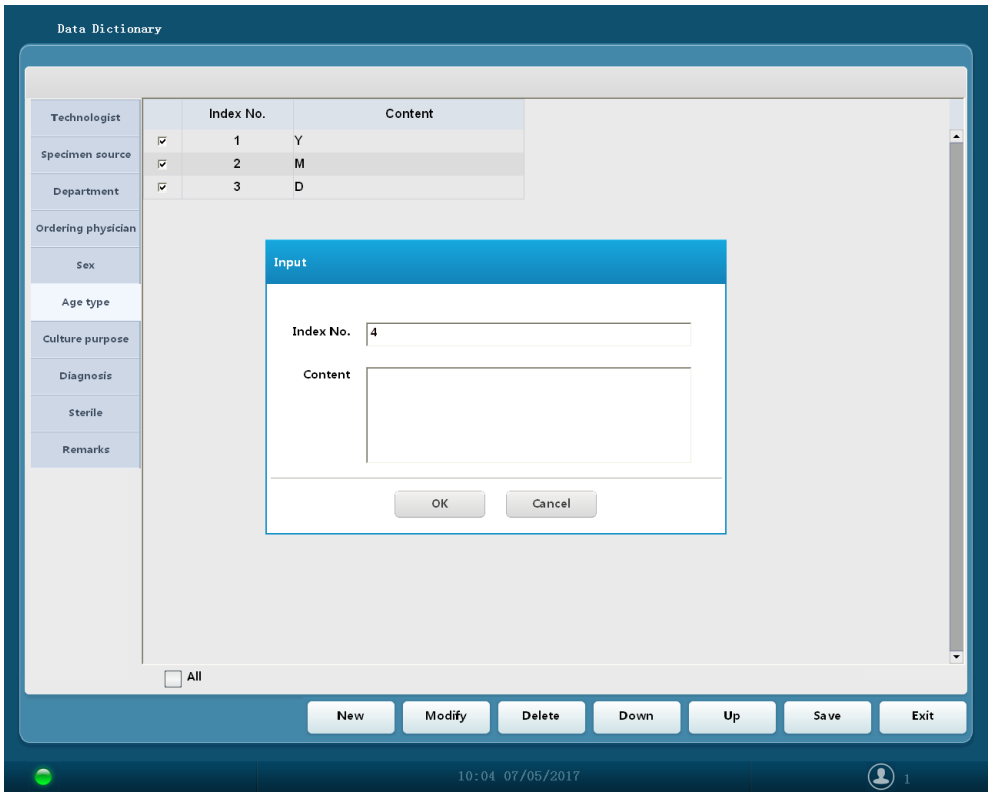


Fig 3-12

3.2.1.7 Culture Purpose Setting

Culture purpose setting: System default setting; Users can add, modify and delete culture purposes;

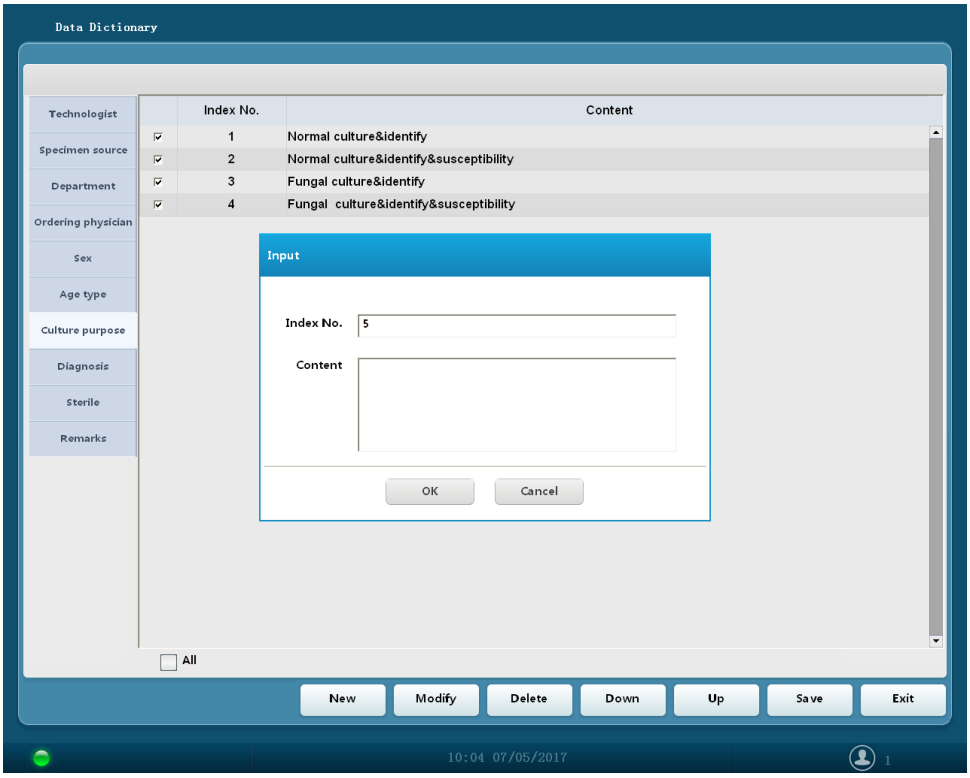


Fig 3–13

3.2.1.8 Diagnosis Setting

Diagnosis setting: System default setting; Users can add, modify and delete diagnosis;

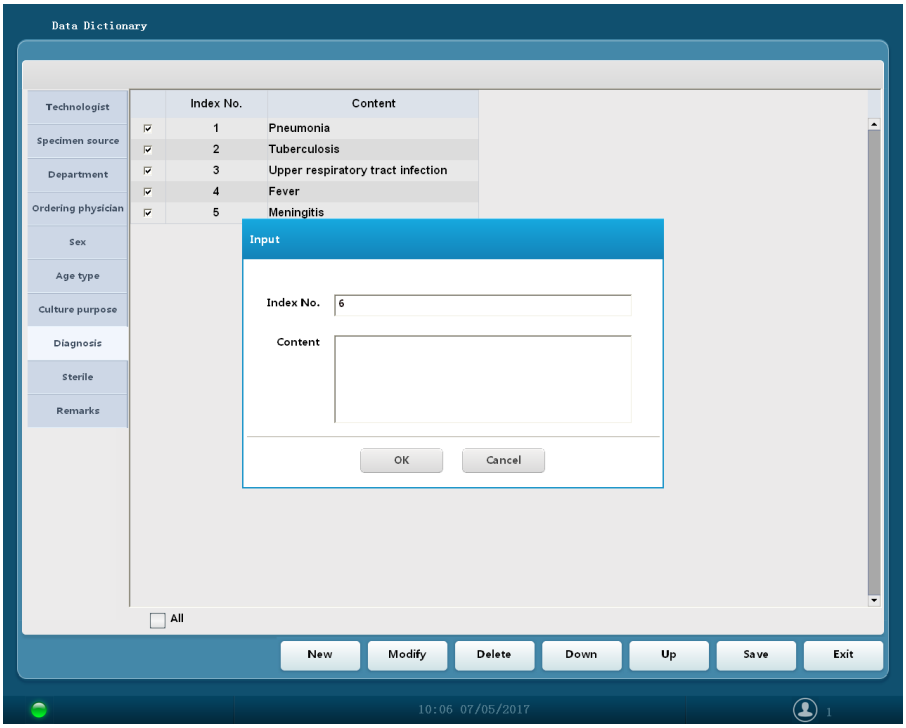


Fig 3-14

3.2.1.9 Sterile Description Setting

Sterile description setting: System default settings;

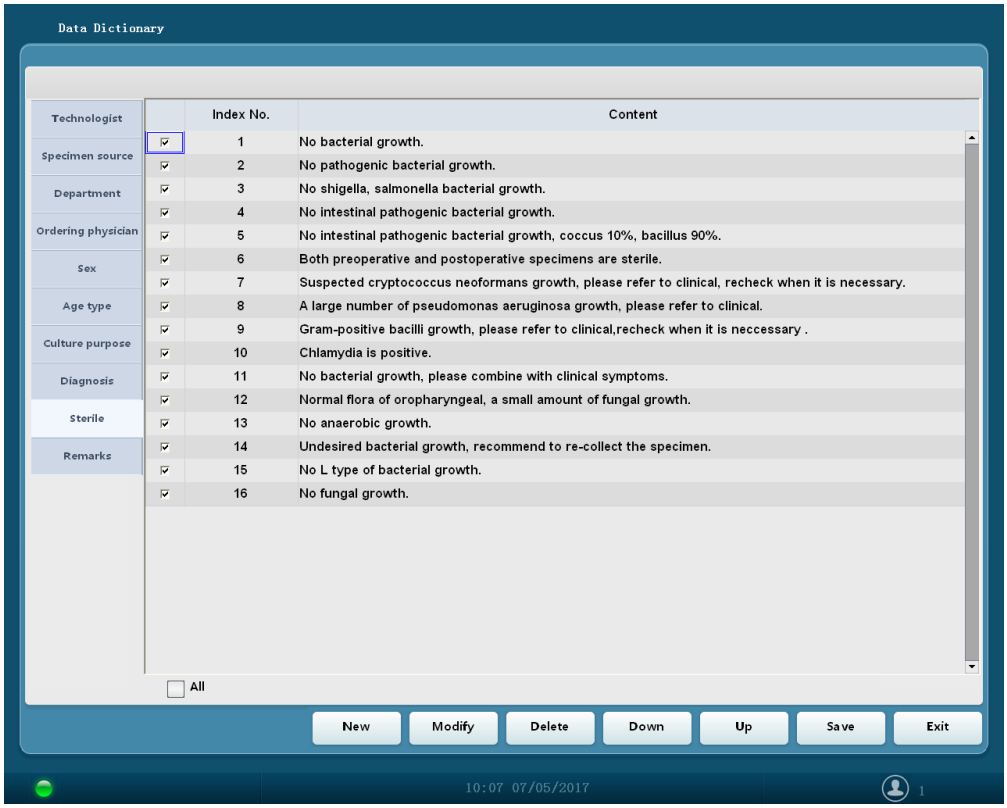


Fig 3–15

- Users can add, modify and delete sterilities;

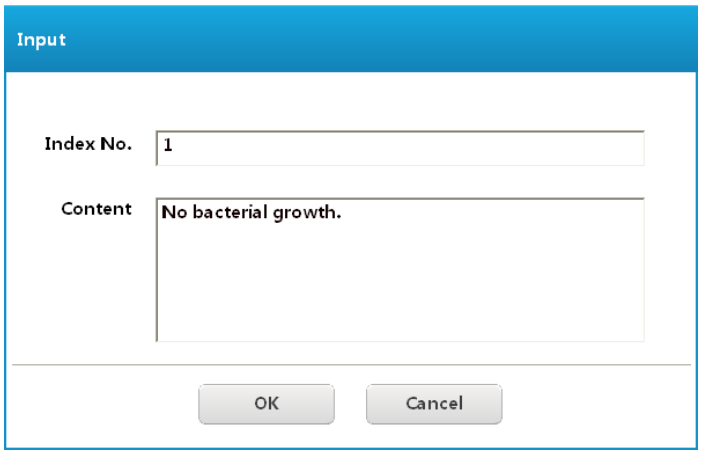


Fig 3–16

3.2.1.10 Remarks Setting

Remarks setting: Users can add, modify and delete remarks. The added content will be displayed in the drop-down list of the test identification module.

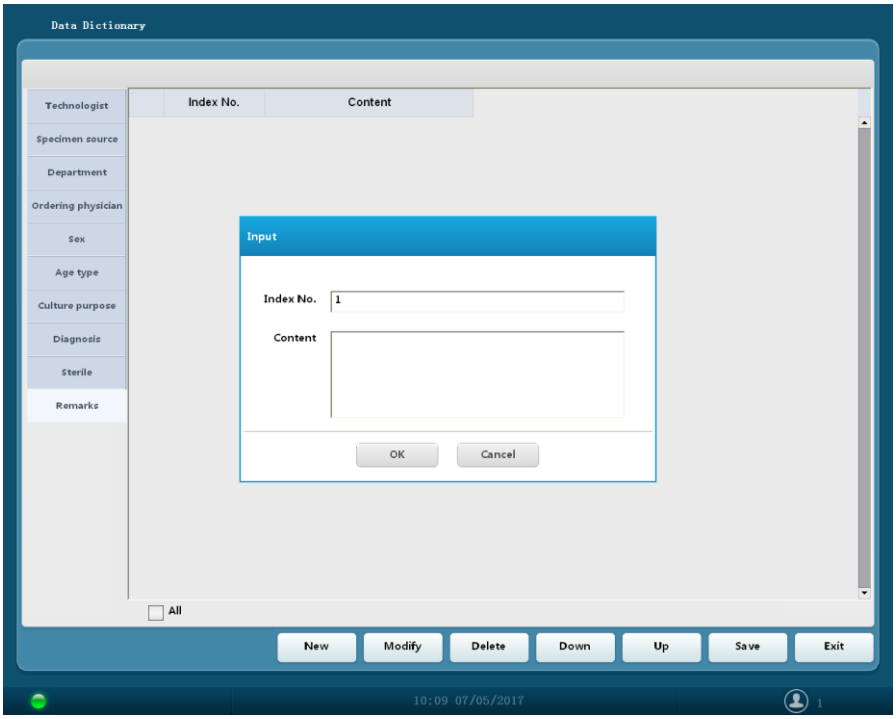


Fig 3-17

3.2.2 Print Setting

Go to “Setting-> Print Set”, the print set interface is showed as below

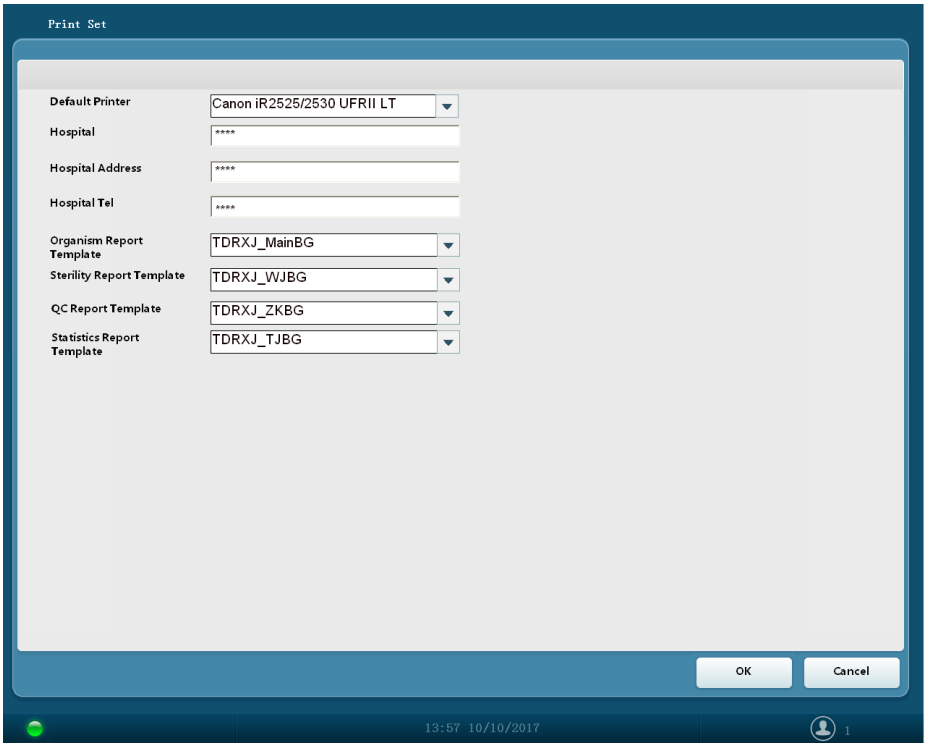


Fig 3-18

- Default printer: Select the installed printer.
- Hospital: Input the hospital name.
- Hospital Address: Input the hospital address.

- Hospital Tel: Input the hospital telephone or microbiological lab telephone.
- Report template: Bacteria report template, Sterility report template, QC report template, and Statistics report template. System has set default settings for them.

3.2.3 Test Card Setting

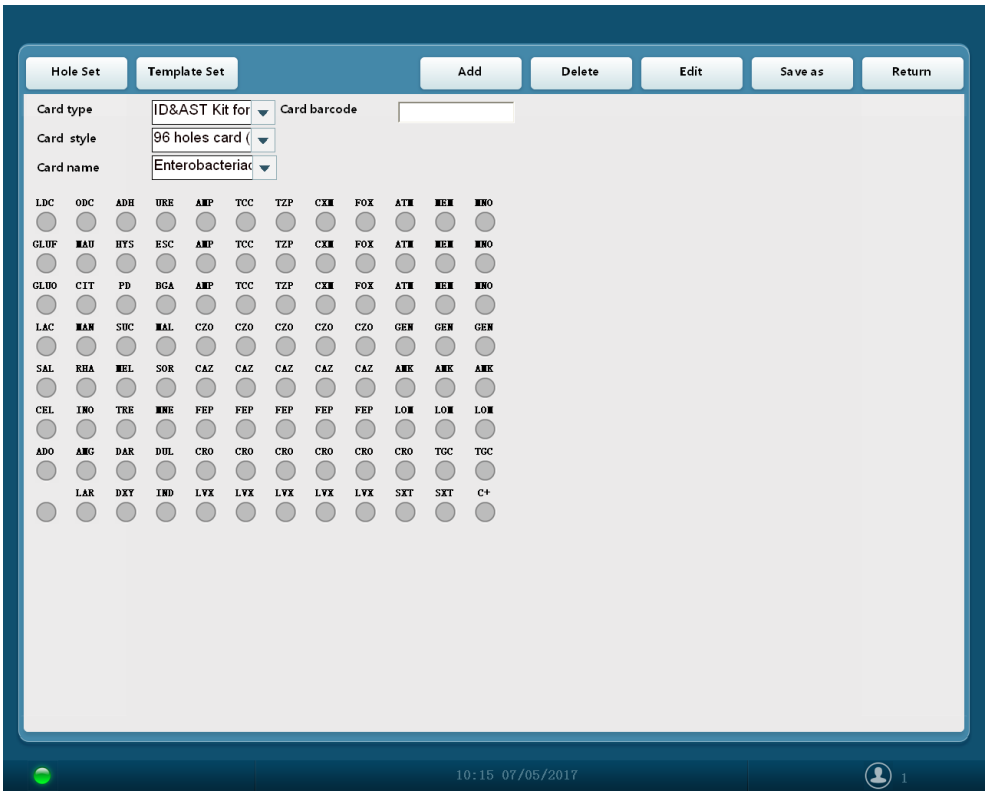


Fig 3-19

Card settings, mainly for:

- For the users who use the customized card, need to adjust or change the biochemical test or susceptibility test;
- According to CLSI drug breakpoint update, need to adjust the concentration of the drugs;
- Bacteria level selection: Users can select the bacteria level according to their own laboratory requirements.

Bacteria are divided into 5 levels:

1、 Level 1 bacteria are clinical common bacteria.

2、 Level 2 bacteria are clinical uncommon bacteria.

3、 Level 3 bacteria are animal-based bacteria.

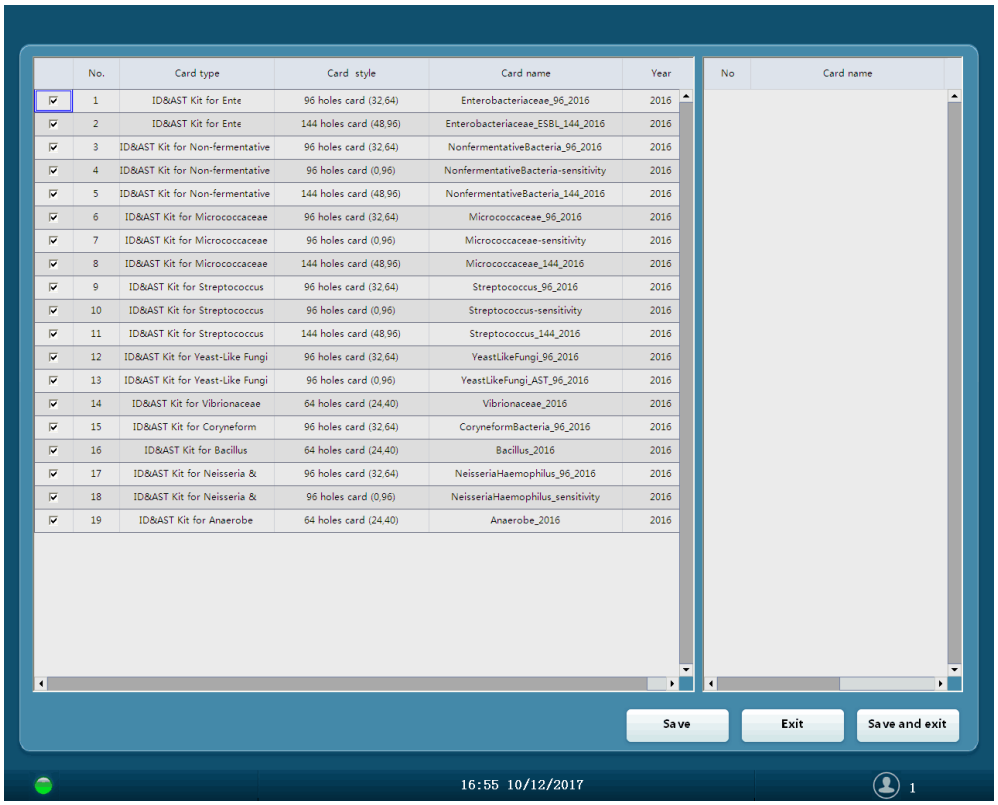
4、 Level 4 bacteria are biological horror bacteria.

5、 Level 5 bacteria are QC bacteria; the results are used for comparison, such as biochemical results are compared with standard results.

Recommendation: Small and medium-sized laboratory: 1 +2 +5, large-scale laboratory: 1 +2 +3 +4 +5.

Template Set:

- According to the card template you are using, select the right test card to do identification. (See below).



	No.	Card type	Card style	Card name	Year
☑	1	ID&AST Kit for Ente	96 holes card (32,64)	Enterobacteriaceae_96_2016	2016
☑	2	ID&AST Kit for Ente	144 holes card (48,96)	Enterobacteriaceae_ESBL_144_2016	2016
☑	3	ID&AST Kit for Non-fermentative	96 holes card (32,64)	NonfermentativeBacteria_96_2016	2016
☑	4	ID&AST Kit for Non-fermentative	96 holes card (0,96)	NonfermentativeBacteria-sensitivity	2016
☑	5	ID&AST Kit for Non-fermentative	144 holes card (48,96)	NonfermentativeBacteria_144_2016	2016
☑	6	ID&AST Kit for Micrococcaceae	96 holes card (32,64)	Micrococcaceae_96_2016	2016
☑	7	ID&AST Kit for Micrococcaceae	96 holes card (0,96)	Micrococcaceae-sensitivity	2016
☑	8	ID&AST Kit for Micrococcaceae	144 holes card (48,96)	Micrococcaceae_144_2016	2016
☑	9	ID&AST Kit for Streptococcus	96 holes card (32,64)	Streptococcus_96_2016	2016
☑	10	ID&AST Kit for Streptococcus	96 holes card (0,96)	Streptococcus-sensitivity	2016
☑	11	ID&AST Kit for Streptococcus	144 holes card (48,96)	Streptococcus_144_2016	2016
☑	12	ID&AST Kit for Yeast-Like Fungi	96 holes card (32,64)	YeastLikeFungi_96_2016	2016
☑	13	ID&AST Kit for Yeast-Like Fungi	96 holes card (0,96)	YeastLikeFungi_AST_96_2016	2016
☑	14	ID&AST Kit for Vibrionaceae	64 holes card (24,40)	Vibrionaceae_2016	2016
☑	15	ID&AST Kit for Coryneform	96 holes card (32,64)	CoryneformBacteria_96_2016	2016
☑	16	ID&AST Kit for Bacillus	64 holes card (24,40)	Bacillus_2016	2016
☑	17	ID&AST Kit for Neisseria &	96 holes card (32,64)	NeisseriaHaemophilus_96_2016	2016
☑	18	ID&AST Kit for Neisseria &	96 holes card (0,96)	NeisseriaHaemophilus_sensitivity	2016
☑	19	ID&AST Kit for Anaerobe	64 holes card (24,40)	Anaerobe_2016	2016

No	Card name

Fig 3-20

3.2.4 Drug Derivation Setting

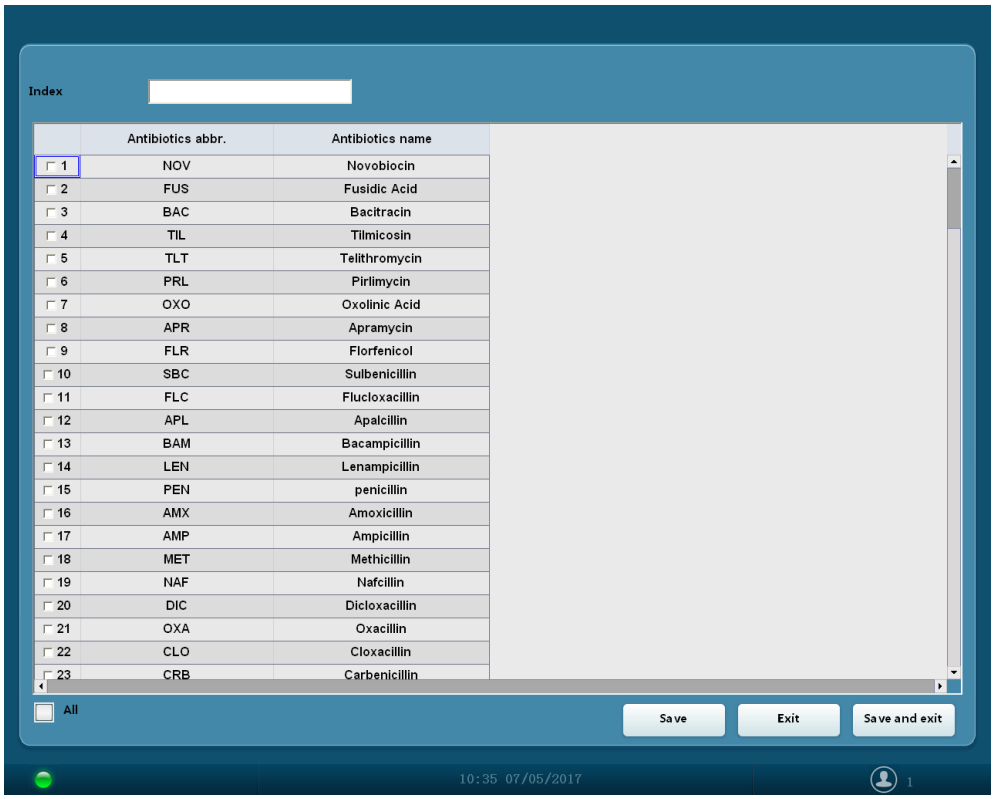


Fig 3-21

Precondition for drug derivation:

Firstly, the derivation rules have been defined in the system.

Secondly, enable the antibiotic based on the hospital's requirements.

Antibiotics derivation setting steps:

1, Check the "√" before the antibiotic option to enable it.

2, Click "Save" or "Save and exit" button.

3.2.5 Patient Information Field Set

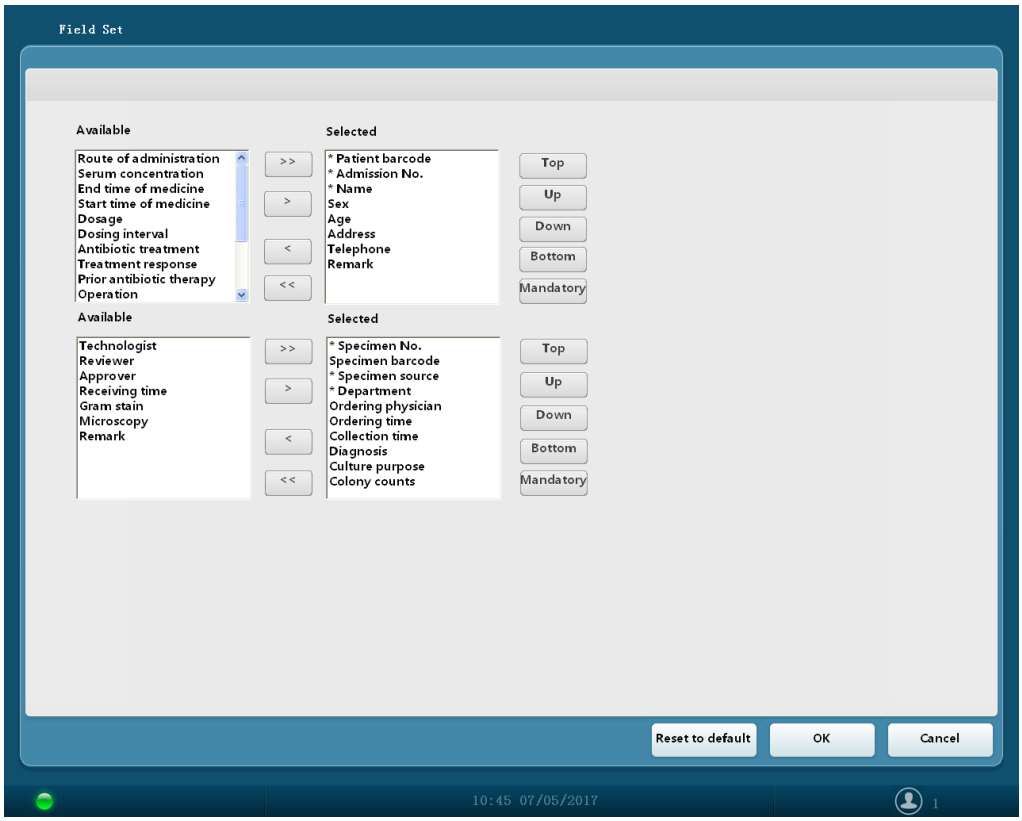


Fig 3–22

- Click “Setting->Patient Info. Field Set” to enter into field set interface.
- If you want to remove the field from selected module, select it and click the “ < ” button;
- If you want to remove all fields from selected module, click the “ << ” button;
- If you want to add the field into selected module, select it and click the “ > ” button;
- If you want to add all fields into selected module, click the “ >> ”button;
- If you want to set the field as mandatory field, select it and click “Mandatory” button, system will mark “*” automatically.
- If you want to set the fields as default settings, click the “ Reset to default ” button;
- According to the requirements of hospital, adjust the field’s position by clicking “Up”, “Down”, “Top” and “Bottom” button.

3.2.6 Query Setting

Query setting contains patient query field set and result query field set;

- Users can use “ >> ” to select all, use “ << ” to remove all, use “ > ” to select individual field and “ < ” to remove individual field.

- According to the requirements of hospital, adjust the field's position by clicking "Up", "Down", "Top" and "Bottom" button.

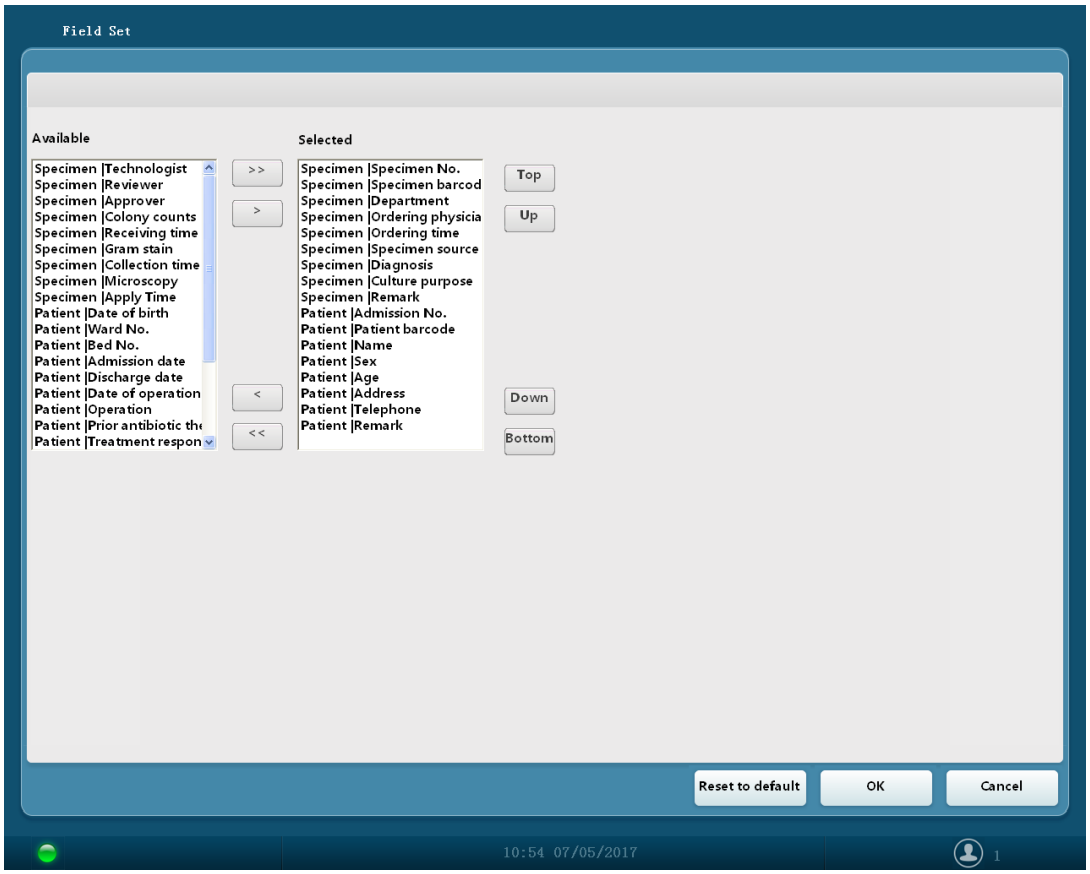


Fig 3–23

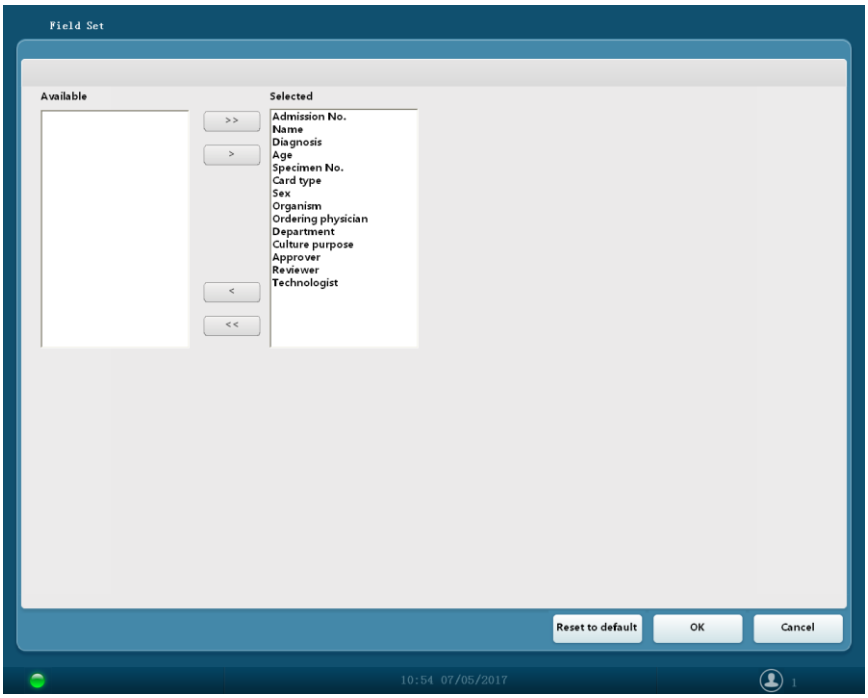


Fig 3–24

3.2.7 User Management

User management is to set the user's permission;

- Password setting：System supports setting a unique login password for each user, select the user, input the password, click “Save”, the password is set successfully;
- Permission Setting: Set different permission for users to achieve the permission classification;

3.2.8 System Setting

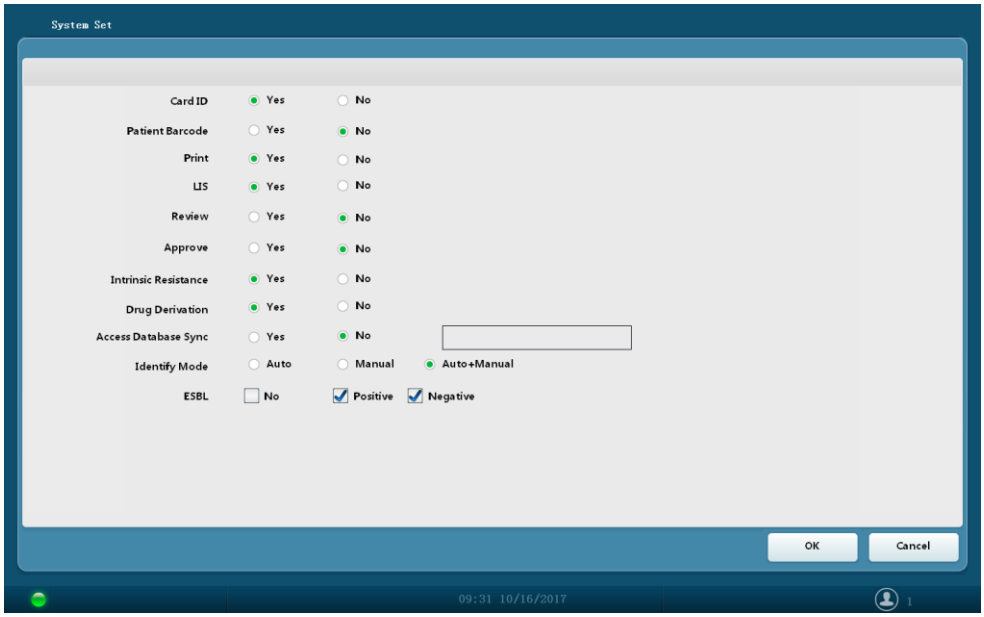


Fig 3–25

System Settings are to enable or disable the functions.

- Enable card barcode function: Check “Yes”, system will identify the card barcode automatically; user does not need to choose the card type manually.
- Enable patient barcode function: Check “Yes”, scan the patient barcode in test module, then press “Enter” button on keyboard, system will match the patient and sample info automatically, user does not need to input the info manually.
- Enable print function: Check “Yes”, the result can be printed, user need to set the print info as described in section 3.2.2.
- Enable LIS function: Check “Yes”, the results will be sent to LIS system.
- Enable review function: Check “Yes”, the test results can be reviewed in the result module.
- Enable approve function: Check “Yes”, the test results can be approved in the result module.
- Enable intrinsic resistance function: Check “Yes”, the expert system will prompt the intrinsic resistance in the results, and modify the report result as “R” automatically.
- Enable drug derivation function: Check “Yes”, the expert system will deduce the antibiotics according to the drug derivation rules.
- Enable access database sync function: Check “Yes”, the system will sync the 300B system results.

- Test mode: Auto mode, Manual mode, Auto + Manual mode.
- ESBL function: Check “No”, system will not report the ESBL result; Check “Positive” and “Negative”, system will report the ESBL results.

3.2.9 Communication Setting

1, Firstly, check the port:

Click “My computer” on the desktop-> Then right click to choose Manage ->Click “Device Manage”->Click “Ports(COM & LPT) to check the port:

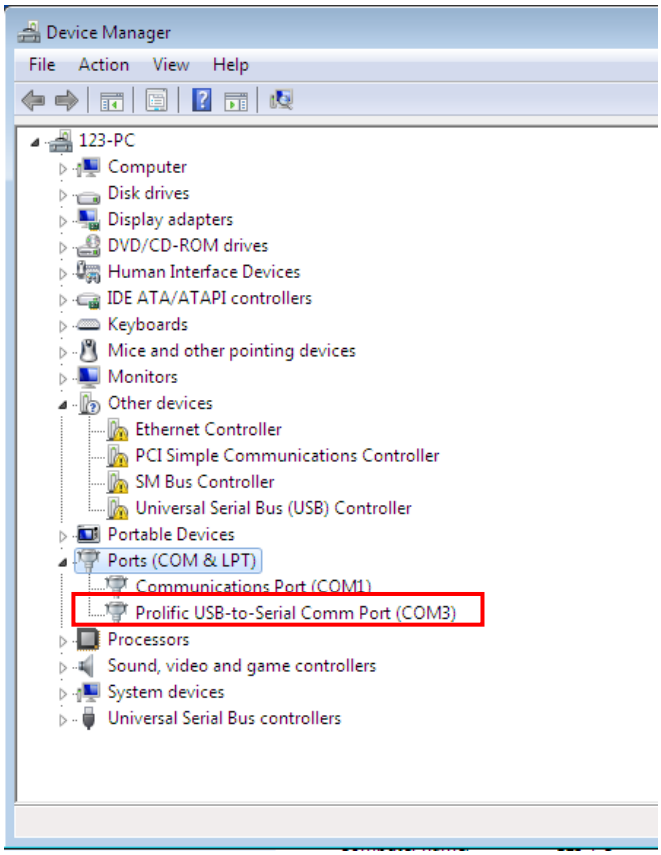


Fig 3-26

2, Go to “Settings”->”Comm Set”, click the drop-down list of serial port to choose the target port;

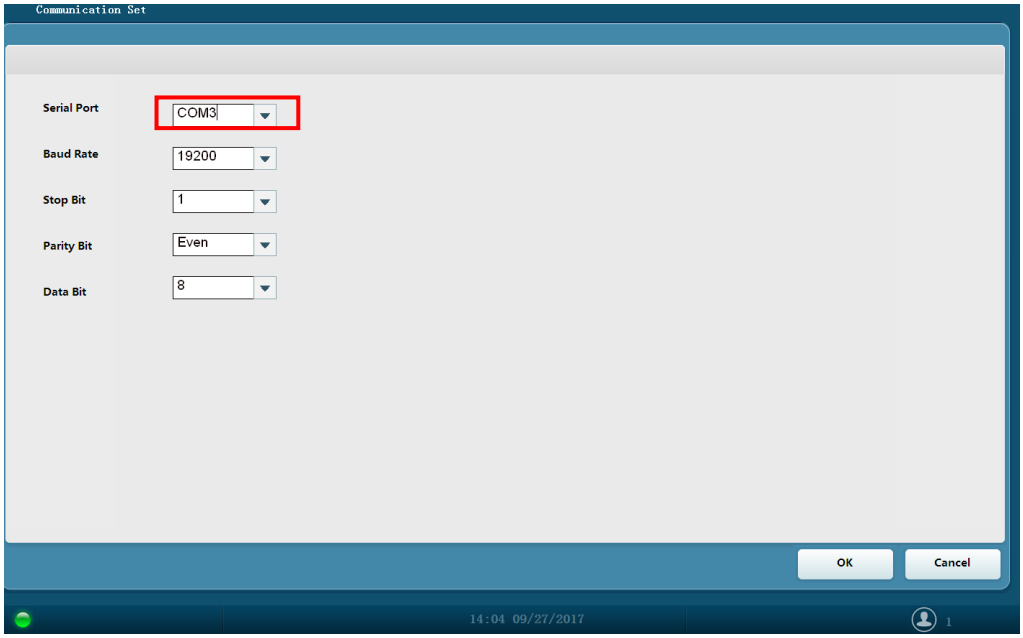


Fig 3-27

3.2.10 LIS Setting

LIS Set: LIS system connection. Please refer to LIS connection guide to get the detailed info.

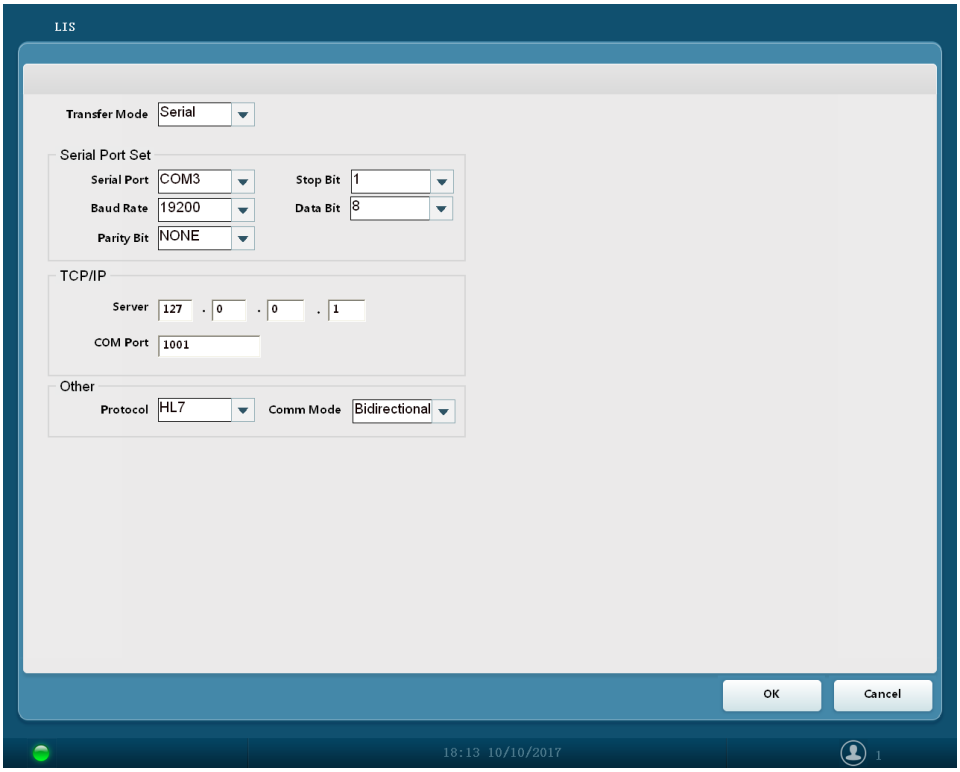


Fig 3-28

3.2.11 Other Settings

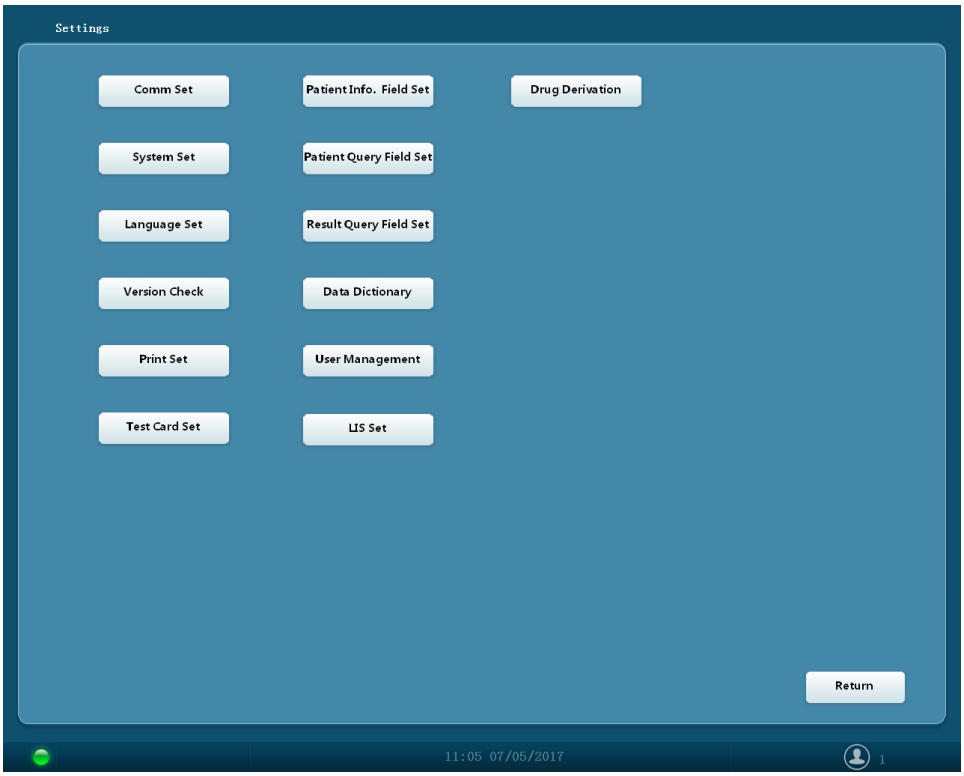


Fig 3–29

Other settings, including language set for changing the software language, version check for checking the version.

3.3 Patient Information

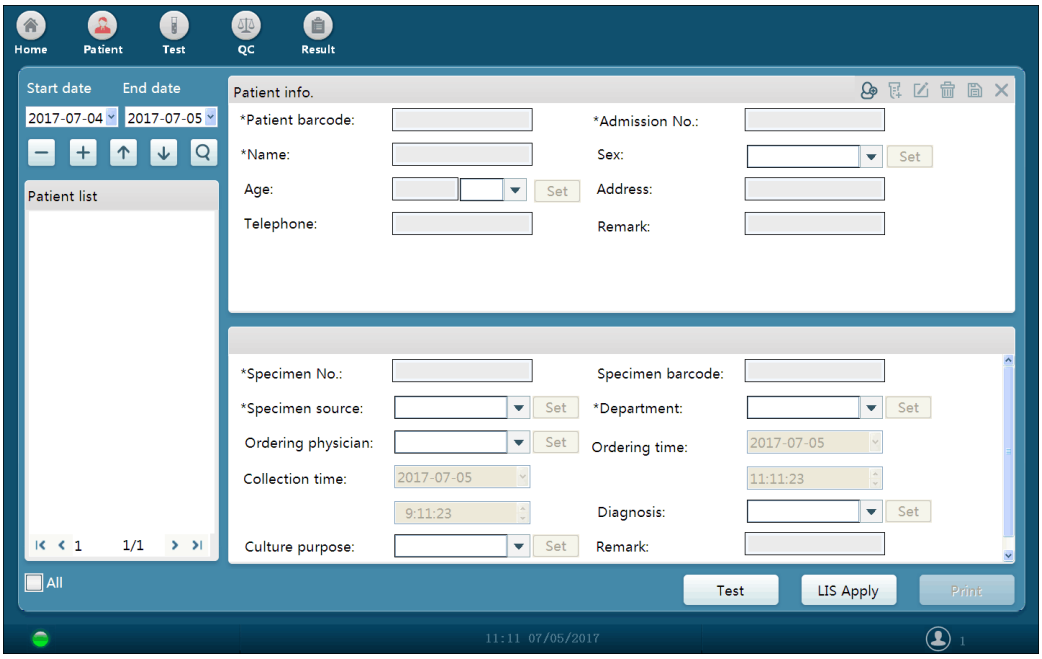


Fig 3–30

- The function keys for adding or editing patient info：

	Add new patient		Add new specimen
	Modify the selected patient or specimen information		Delete the selected patient information
	Save the patient information		Cancel the operation

Note：The function buttons are consistent in other interface;

- The function keys for sorting and querying：

	Collapse all information		Expand all information
	Ascending all information		Descending all information
	Query information		

- The function keys for test, LIS apply and print:：

Test | LIS apply | Print

Test	Go to the test module	LIS apply	Download the patient info from LIS
Print	Print the patient information and test results		

3.3.1 Add New Patient

- Procedure：

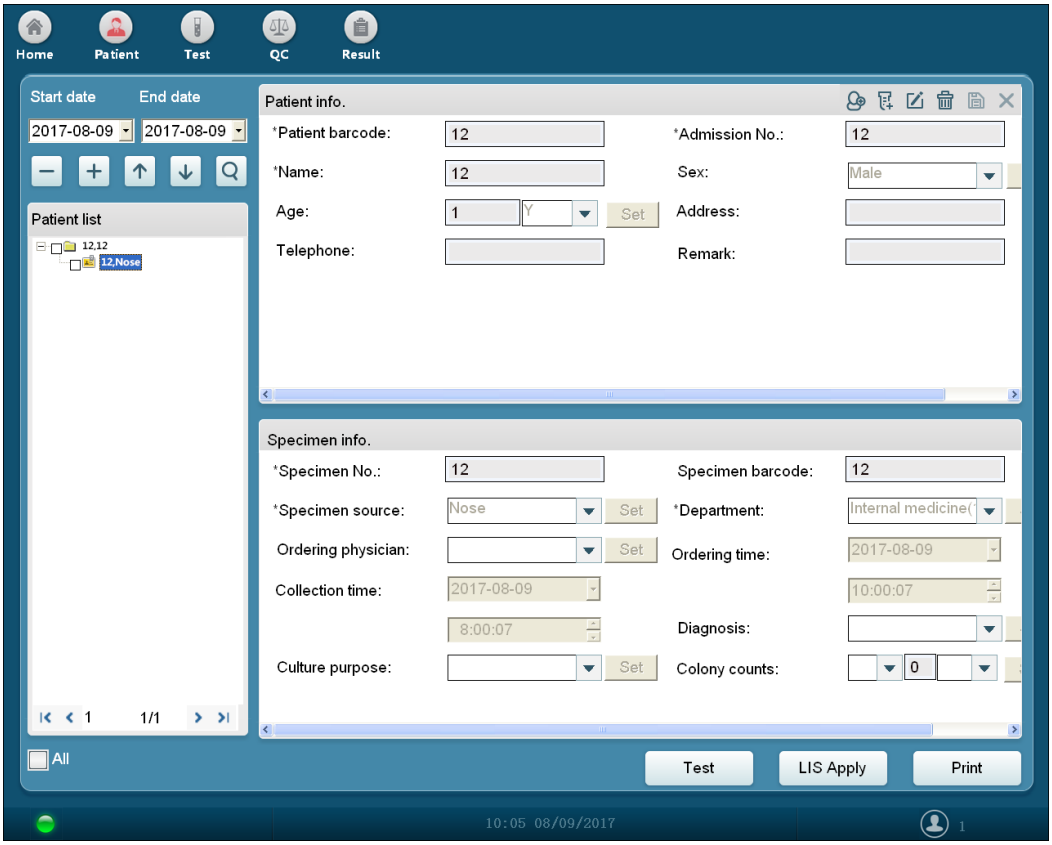


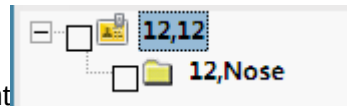
Fig 3-31

1. Go to “Patient” module, click“ ”,input the patient information. Note: The fields with “*” are mandatory fields;

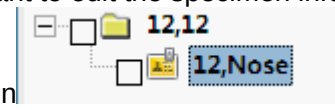
2. Input the Name, Specimen Source, Culture Purpose etc. You can use mouse or the enter key on the keyboard to control your input. Click the set button on the right of fields to enter the corresponding dictionary, and then you can add, delete or modify the drop-down list fields.

3. Click the “ ”to save the patient information, you can continue to add the next patient.

4. If you want to add multiple specimens for the patient, select the patient, click “ ”to add new specimen.

5. If you want to edit the patient information, select the patient 

, click “ ” to edit the patient information.

6. If you want to edit the specimen information, select the specimen 

, click “ ” to edit the specimen information.

7. If you want to view a patient or specimen information, select the patient or specimen, the right interface will display all relevant information.

3.3.2 Query Patient Information

Procedure：

1. Click “ ”,pop out the querying interface, as the Fig 3–32;
2. Enter or select the relevant value in the input line;
3. Click “OK”, the system search the matched records and list them out;

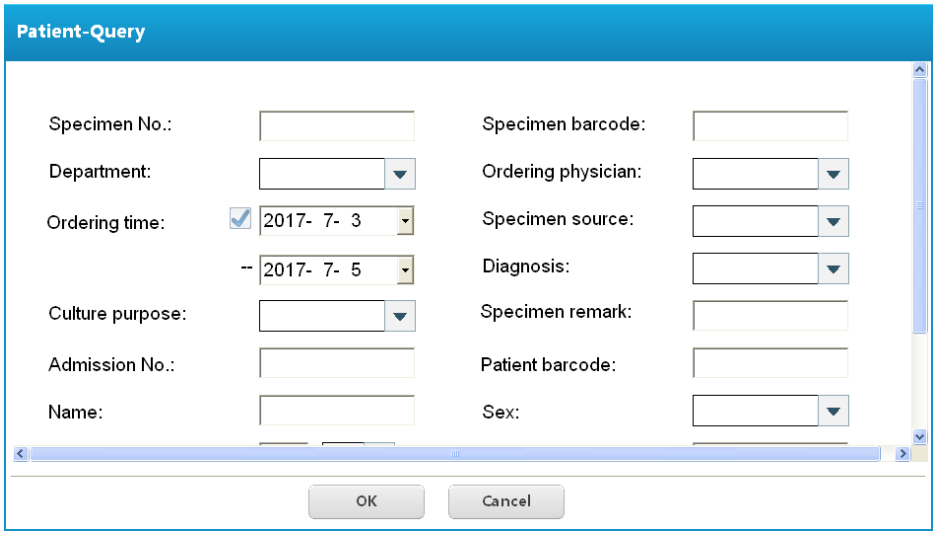


Fig 3–32

3.4 Test

Enter into the test module as below; choose the untested patients to do test.

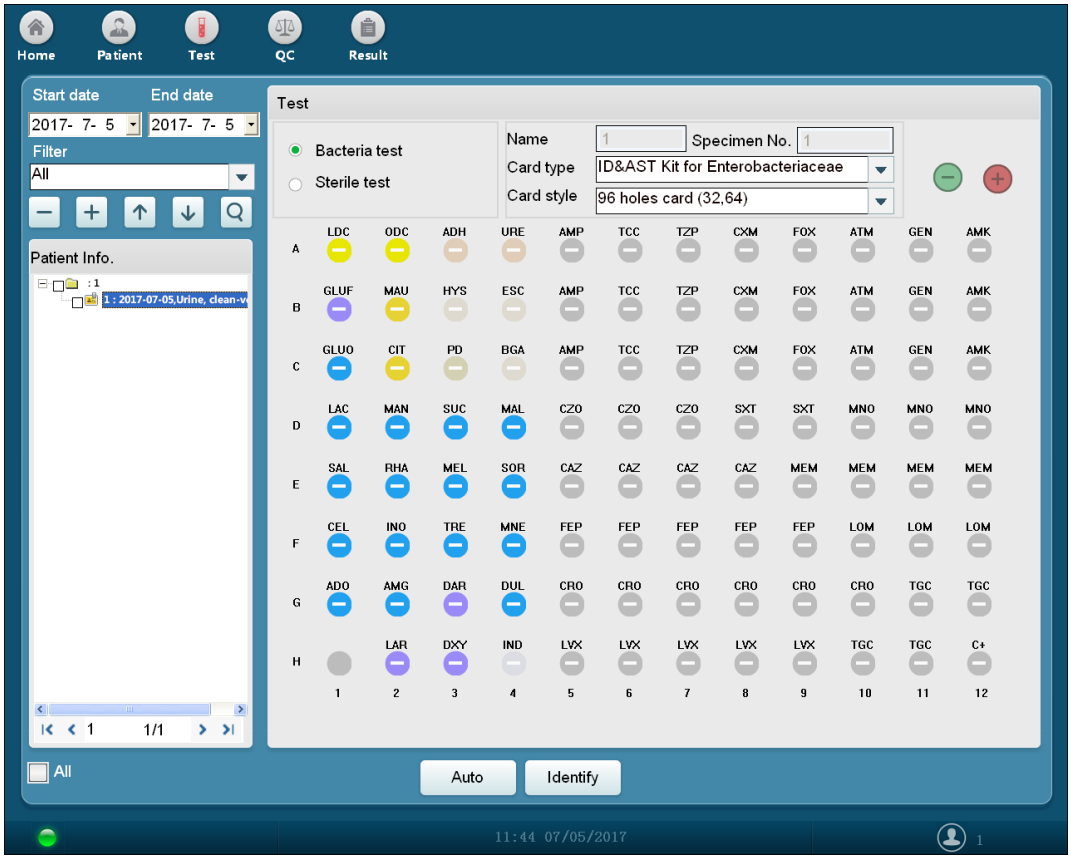
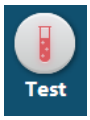
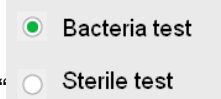


Fig 3–33

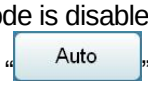
3.4.1 Bacteria Test

3.4.1.1 Automatic Mode

1. Refer to section 3.2.8, enable the test mode as automatic mode;
2. Click ,  ,

to enter the test module;
3. The default test type is bacteria test“ 

”;
4. Select the untested patients to do test, specific steps are described as below:：
   a) If the card barcode is enabled, click “

” to start identifying;
   b) If the card barcode is disabled, user need to select the card type manually(See Fig 3-34), then click “ 

” to start identifying;

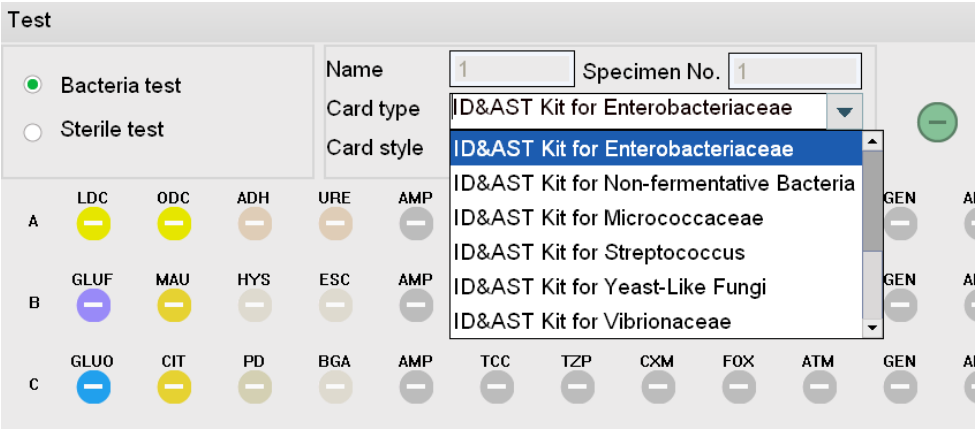


Fig 3–34

5. System analyze the tests automatically and give the results：

   a) If there is uncertain biochemical result(s) and the test affect the organism analysis, the system will pop out a dialog box (see Fig 3-35), user need to take out the card from the instrument and confirm the test result manually.

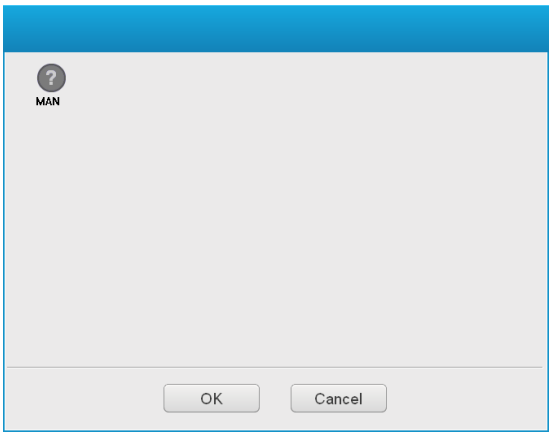


Fig 3–35

   b) If there is no uncertain biochemical or susceptibility results, the system will give the identification results and expert prompts.(See Fig 3-36),

   c) If the system give a number of results (See Fig 3-37), user should combine the identification rate, similarity, unusual experiment prompt, specimen source, colony morphology and staining characteristics to analyze and select the appropriate identification results. Click “ OK ”, system give the final result and expert prompts.

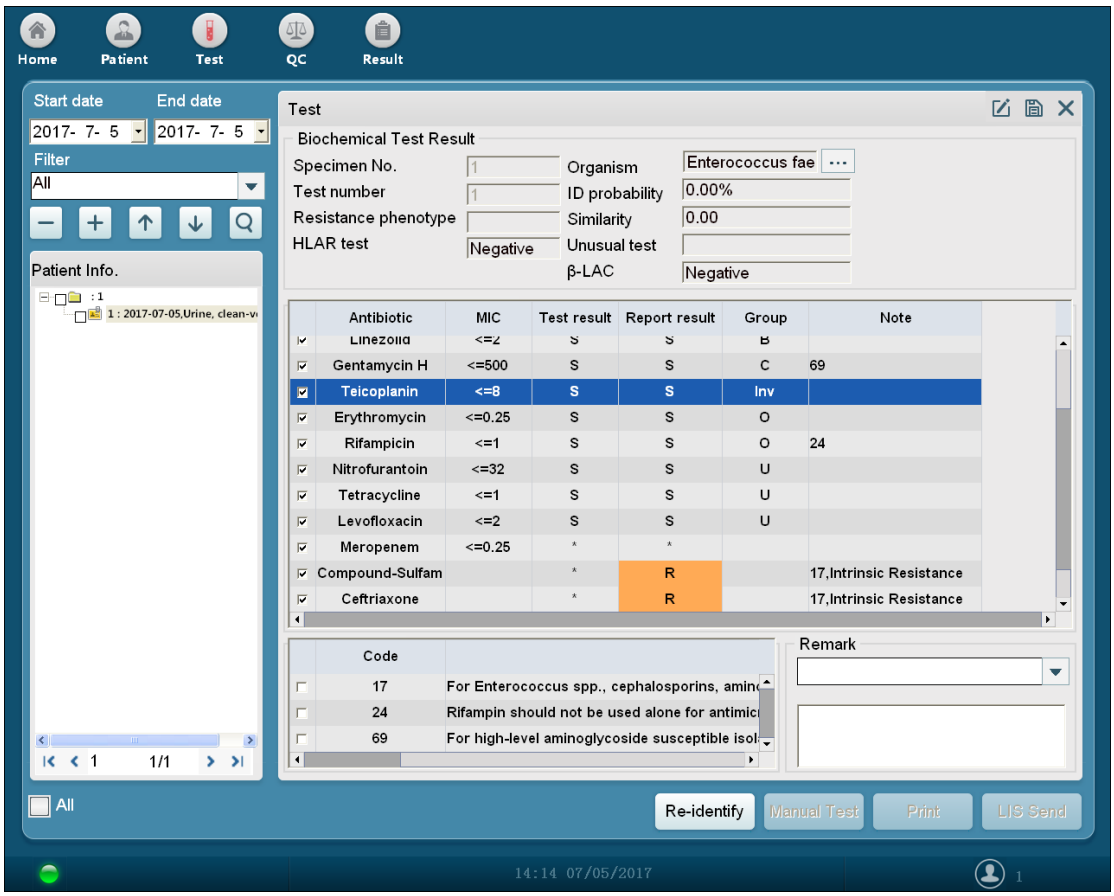


Fig 3–36

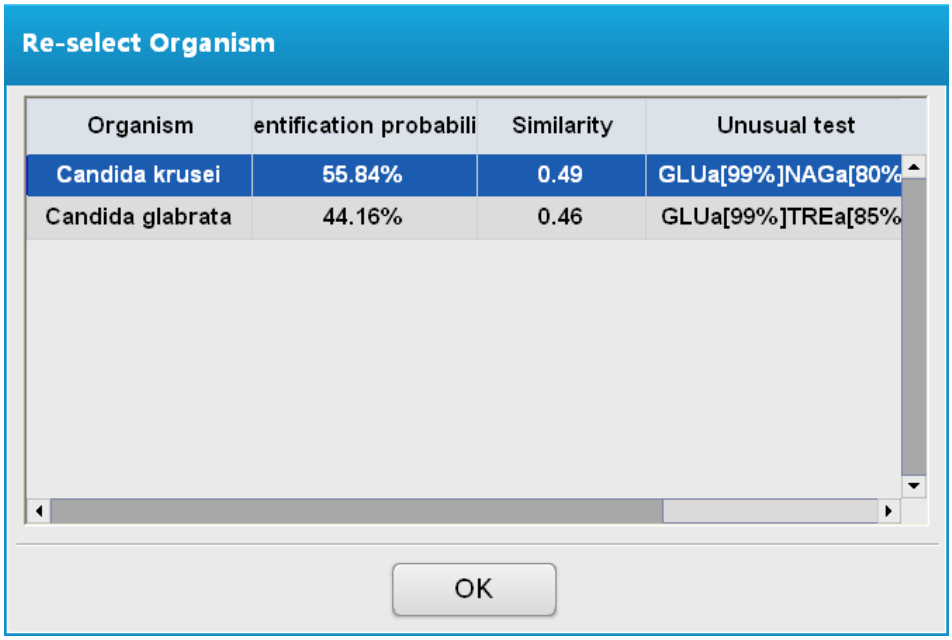


Fig 3–37

6. Check the test result, if it is correct, click“ ”to save the result, test is completed.
7. If you are suspicious of test results, need to re-do the identification, you can click “Re-identify” to return to test interface and modify the test results manually.(： Positive, ： Uncertain, ： Negative)

8. Then click "Identify", the system will give the identification results and expert prompts. (See Figure 3-36).

9. Click" "to save the result, test is completed.

10. Click "Print", system will go to the print preview interface as below figure:

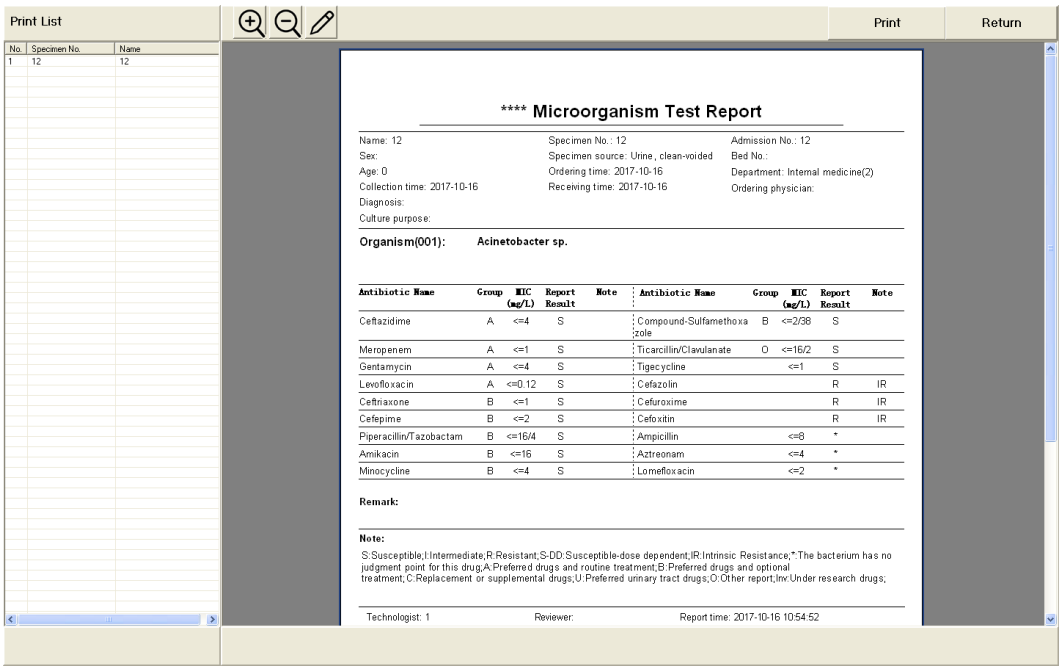


Fig 3–38

11. Click "", it will zoom in the report; click "", it will zoom out the report; click "", it will go to the report template interface, you can adjust the template, then save it to take effect. Click "Print" button to print the report.

3.4.1.2 Manual Mode

1. Refer to section 3.2.8, enable the test mode as manual mode;

2. Click, Test, Test to enter the test module;

3. The default test type is bacteria test "Bacteria test / Sterile test";

4. Select the untested patients in left interface, select the card type in right interface;

5. According to the test card result, enter the biochemical and susceptibility test results to system manually.(See Fig 3-33):

   a) Biochemical tests：If the result is positive, click on the system hole as "⊕"; If the result is uncertain, click on the system hole as"?";If the result is negative, click on the system hole as"⊖";

b) Susceptibility tests：If the hole is turbid or particle growth, click on the system hole as“ ”(Resistant);If the hole is clear, click on the system hole as“ ”(Susceptible);

6. After you entering all results, click “Identify”, system will analyze and give the final identification results.
7. If you are suspicious of the identification results, refer to section 3.4.1.1 Steps 7-9;
8. Click“Print”, system will go to the print preview interface as below figure:

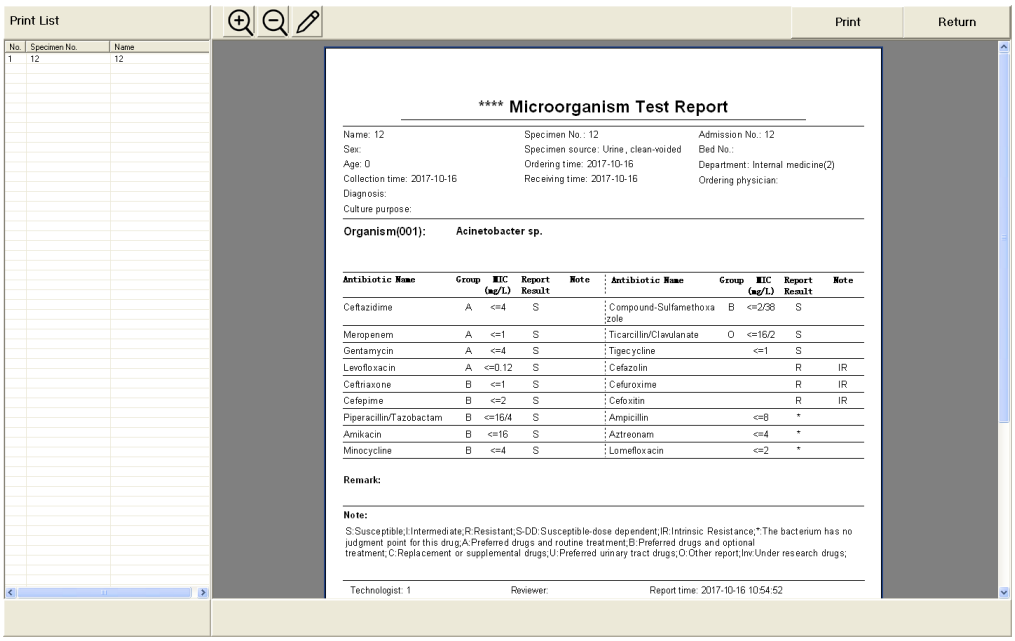
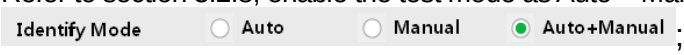


Fig 3-39

9. Click “ ”, it will zoom in the report; click “ ”, it will zoom out the report; click “ ”, it will go to the report template interface, you can adjust the template, then save it to take effect. Click “Print” button to print the report.

3.4.1.3 Automatic + Manual Test Mode

1. Refer to section 3.2.8, enable the test mode as Auto + Manual Mode, see figure below;



;

2. You can select any of test modes; if you select Auto mode, please refer to section 3.4.1.1; if you select Manual mode, please refer to section 3.4.1.2.

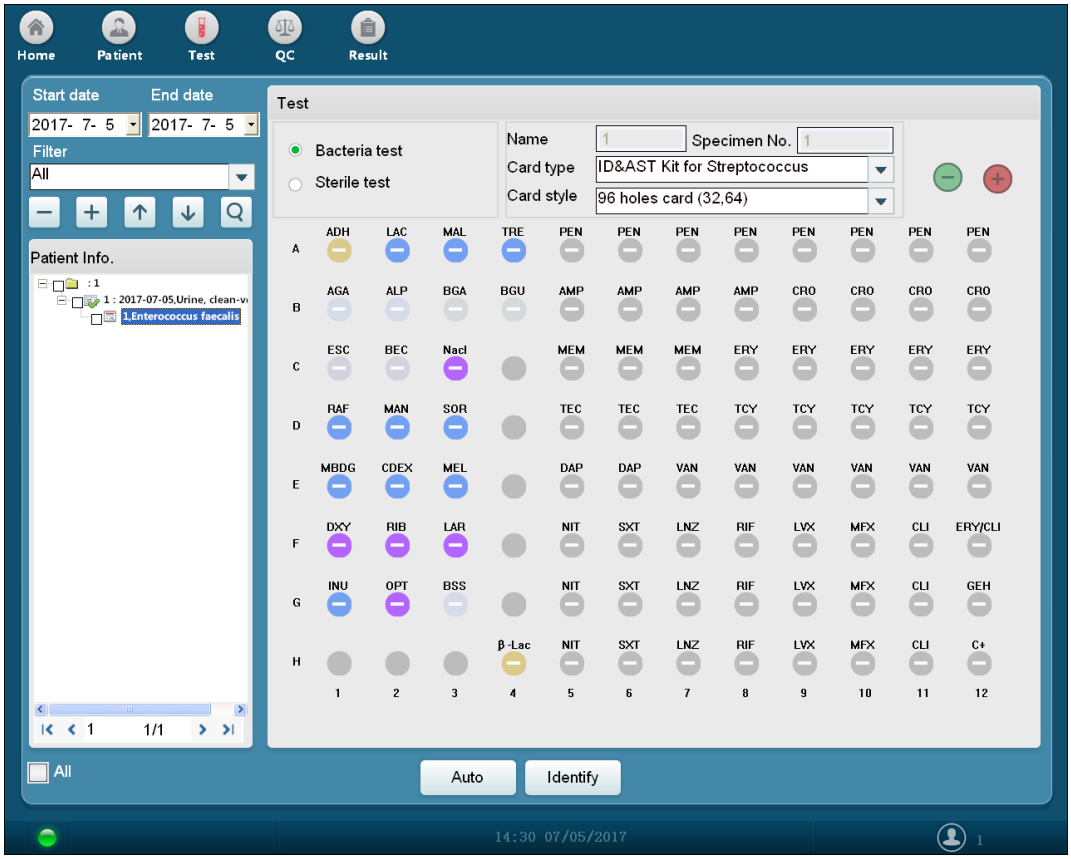


Fig 3-40

3.4.1.4 Antibiotics derivation

For some bacteria, if two or more of the efficacy of antibiotics are equivalent, system will prompt the antibiotics derivation. For example, the Enterobacteriaceae bacteria, Resistance / susceptibility of Ceftriaxone and Cefotaxime are equivalent. (See Figure 3-41).

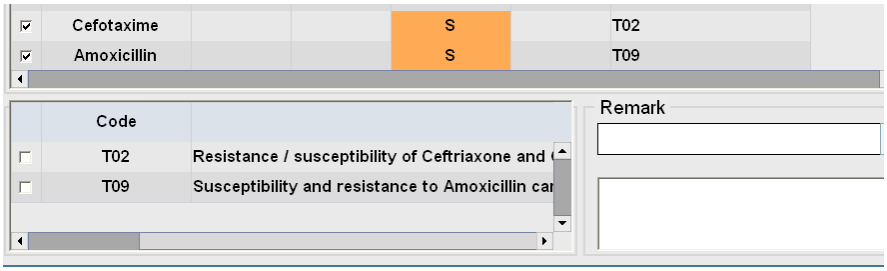


Fig 3–41

3.4.1.5 Intrinsic Resistance

If a bacterium is intrinsic resistant to one or more antibiotics, the susceptibility test result is "S", system will change the report result as "R" automatically and give the prompt" Intrinsic Resistance" in note field. (See Fig3-42)

	Antibiotic	MIC	Test result	Report result	Group	Note
☑	Teicoplanin	<=8	S	S	inv	
☑	Erythromycin	<=0.25	S	S	O	
☑	Rifampicin	<=1	S	S	O	24
☑	Nitrofurantoin	<=32	S	*	U	
☑	Tetracycline	<=1	S	*	U	
☑	Levofloxacin	<=2	S	*	U	
☑	Meropenem	<=0.25	*	*		
☑	Compound-Sulfam		*	R		17,Intrinsic Resistance
☑	Ceftriaxone		*	R		17,Intrinsic Resistance
☑	Moxifloxacin	<=1	*	*		
☑	Clindamycin		*	R		17,Intrinsic Resistance

Fig 3–42

3.4.1.6 Manual Antibiotics

After saving the test results, if you want to add manual antibiotics results, please refer to below:

1. After saving the test results, click“ Manual Test ”,enter into the manual antibiotics interface.（See Fig 3–43）
2. Input the part or complete antibiotic name in the index input line, system will search the matched antibiotic(s) and list them out, select the target antibiotic, the selected antibiotic will display on the left, input the test value(MIC or K-B diameter), and select the test result(S,I,R,S-DD), then click " Add ", the antibiotic is added successfully;
3. If the results of manual antibiotics are added incorrectly, or if the added result needs to be deleted, you can modify and delete the results by the following function key " Delete Modify Save Cancel ";
4. If the operations of manual antibiotics are completed, click“ ✕ ”to exit the manual antibiotic interface;
5. The added manual antibiotics can be viewed at “Result->Manual Antibiotics”;

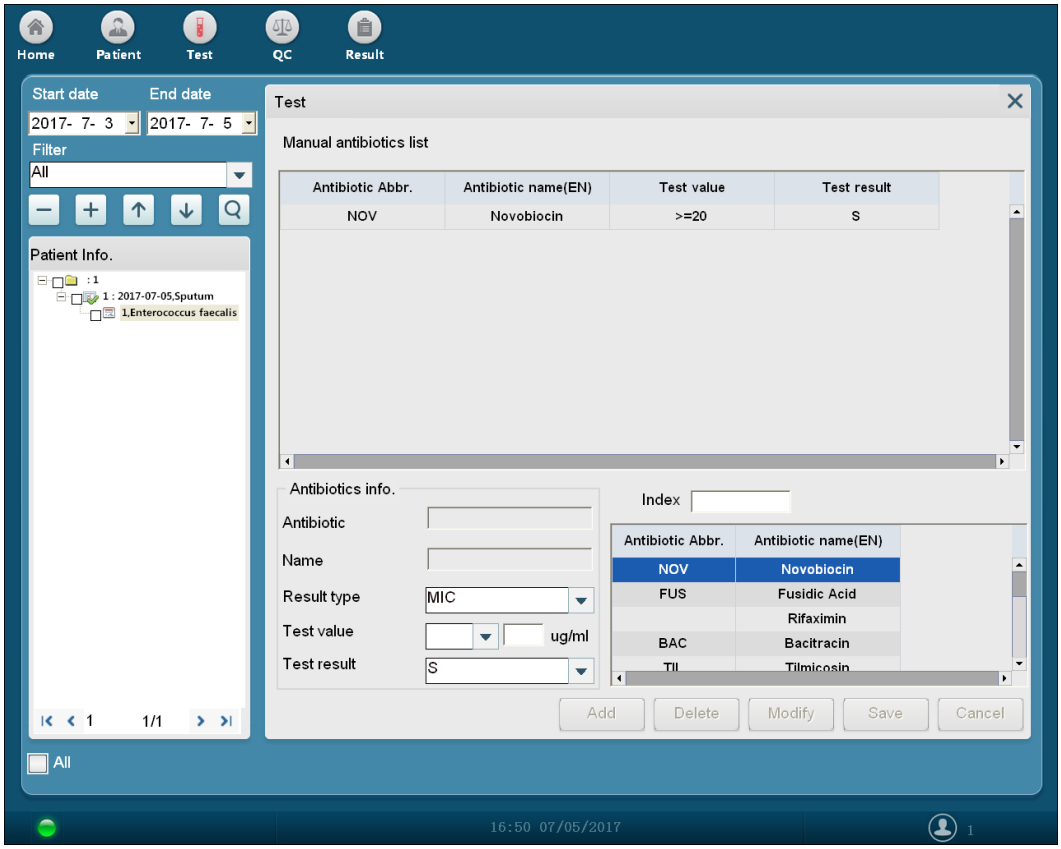
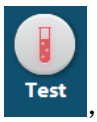
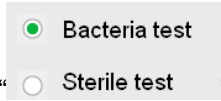


Fig 3–43

Note：Expert system does not apply to manual antibiotics.

3.4.2 Sterile Test

Procedure：

1. Click ,  , to enter the test module,the default test type is bacteria test “  ”;
2. Select the untested patients in left interface, click “Sterile test”, go to the sterile test interface.(See Fig 3-44);
3. Check the target items in the sterile list. Multiple selections are allowed.

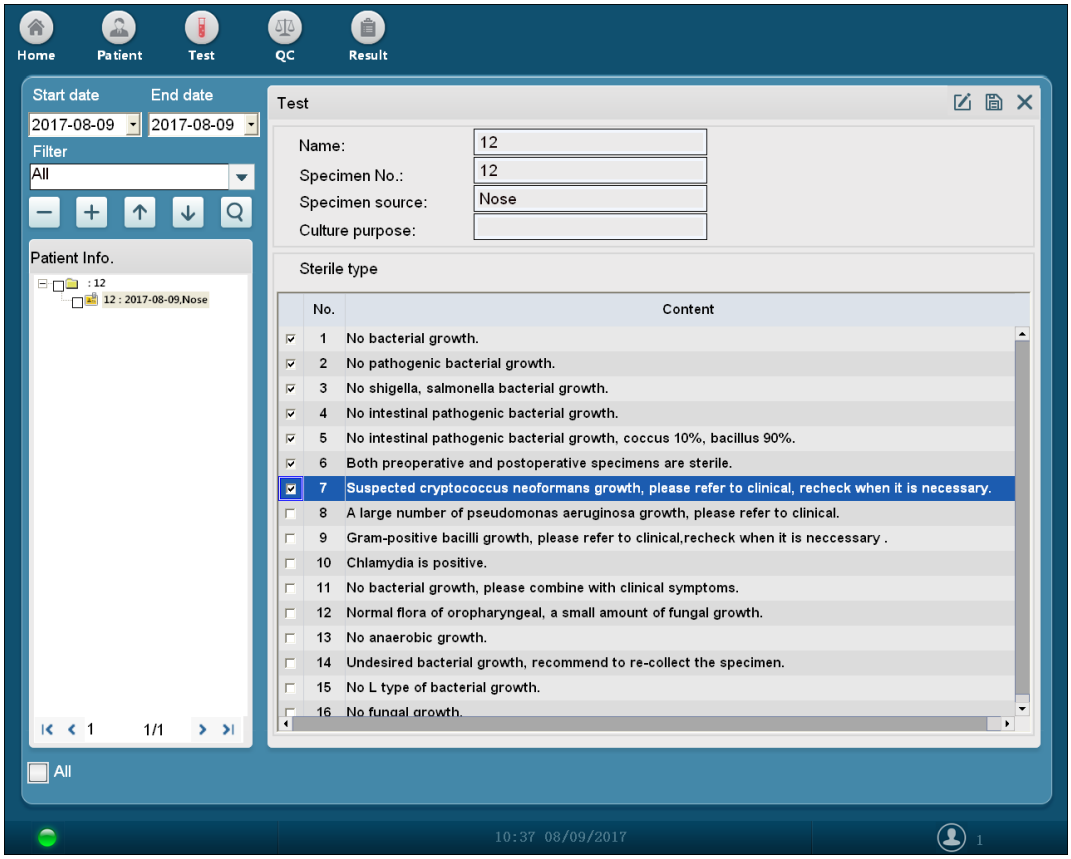


Fig 3–44

4. Click Save. Click Print, system will go to the print preview interface as below figure:

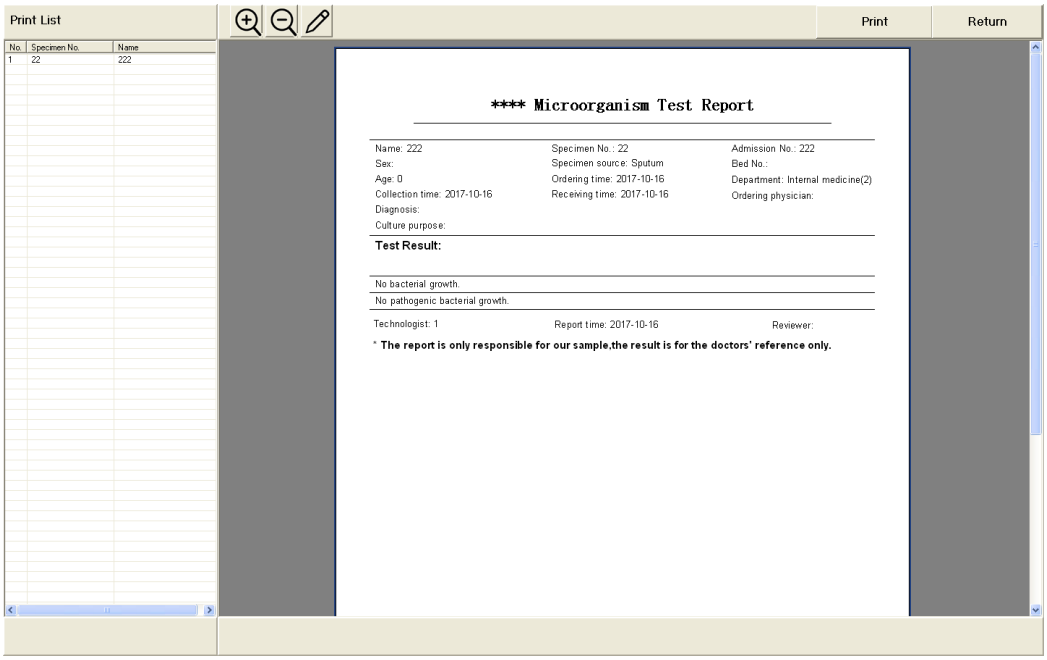


Fig 3-45

5. Click “ ”, it will zoom in the report; click “ ”, it will zoom out the report; click “ ”, it will go to the report template interface, you can adjust the template, then save it to take effect. Click “Print” button to print the report.

3.4.3 QC Test

Procedure:

1. Click “QC”; enter into the QC test interface.

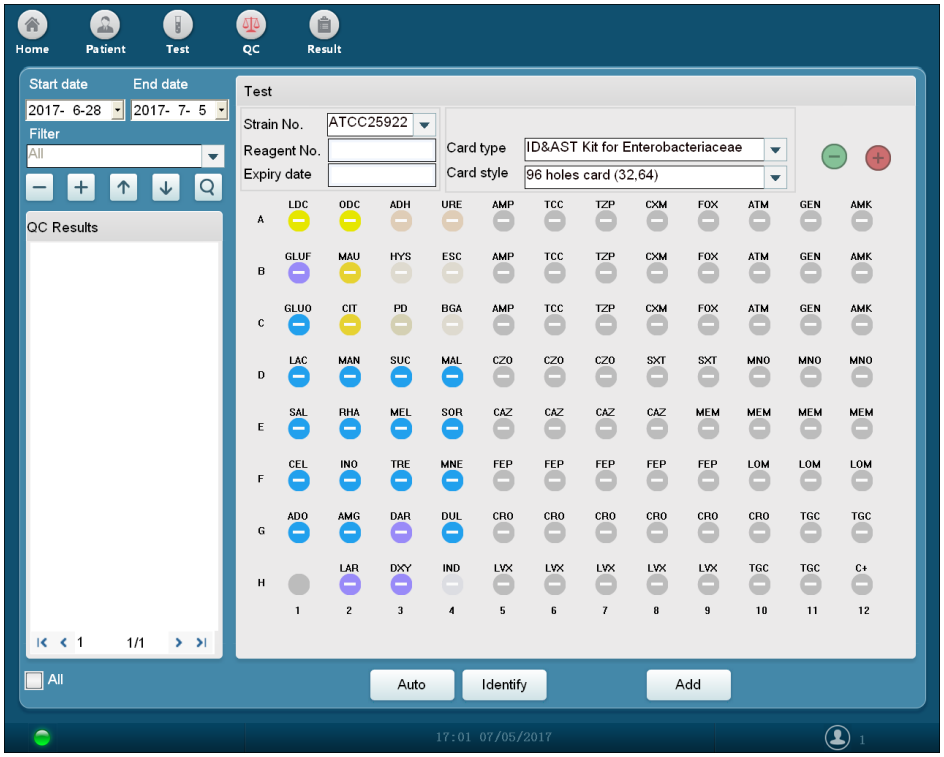


Fig 3–46

2. For intra-Laboratory QC test, pull down the strain No. list to select the target ATCC strain No.

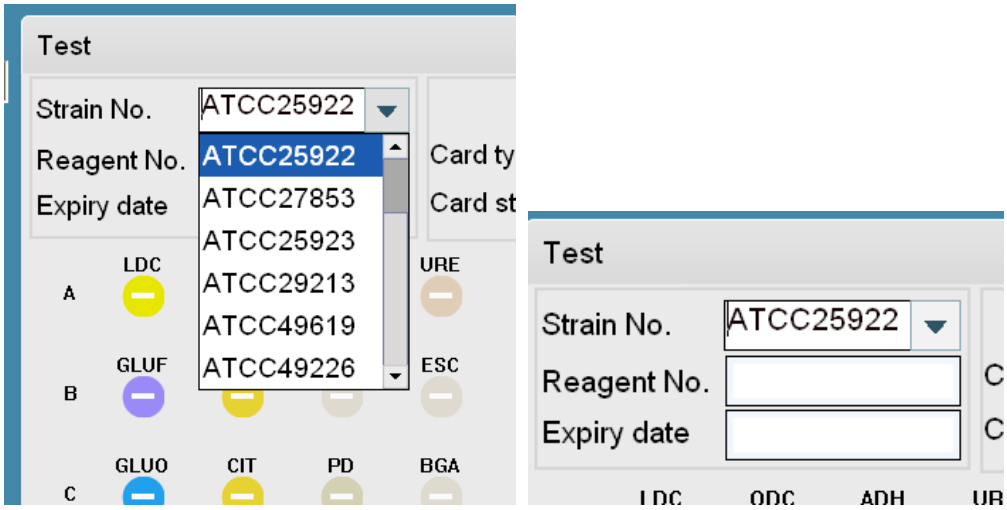


Fig 3–47

3. Enter the factory lot number of test card in reagent No. field, enter the expiry date of the card in expiry date field, select the matched QC card type;

a) If the test mode is “Auto”, click “Auto” to start identifying; System analyze the tests automatically and give the results;(See Fig 3-48)

b) If the test mode is “Manual”, enter the biochemical and susceptibility test results to system manually, click“Identify”,system analyze the tests automatically and give the results;（See Fig 3–48）

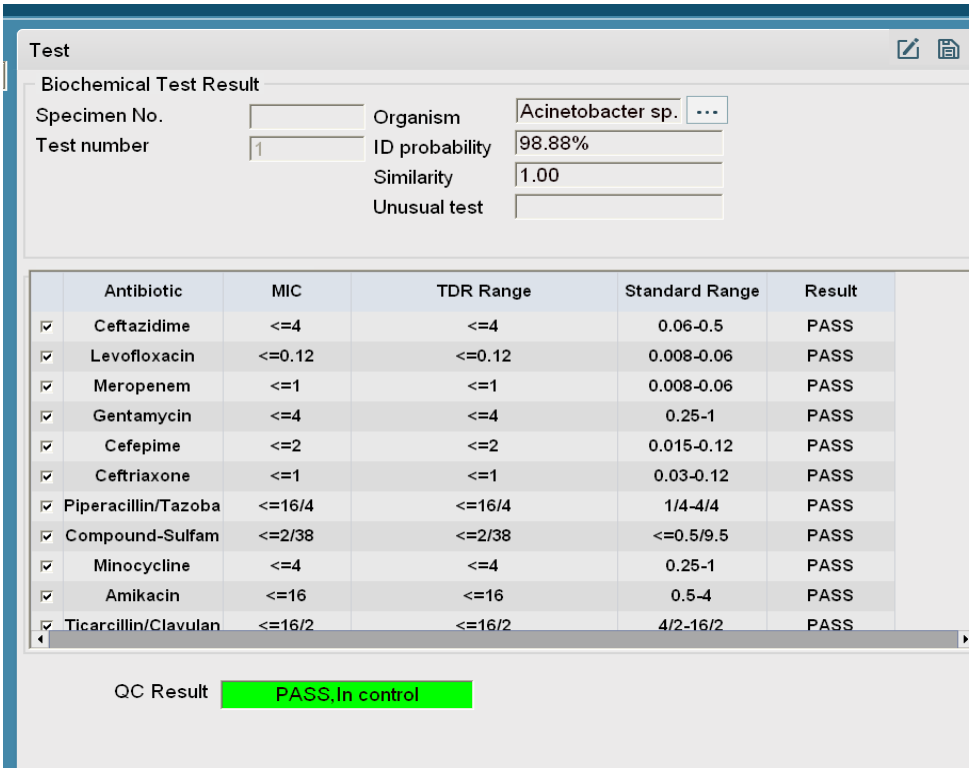


Fig 3–48

4. Click “Print” ,system will go to the print preview interface as below figure:

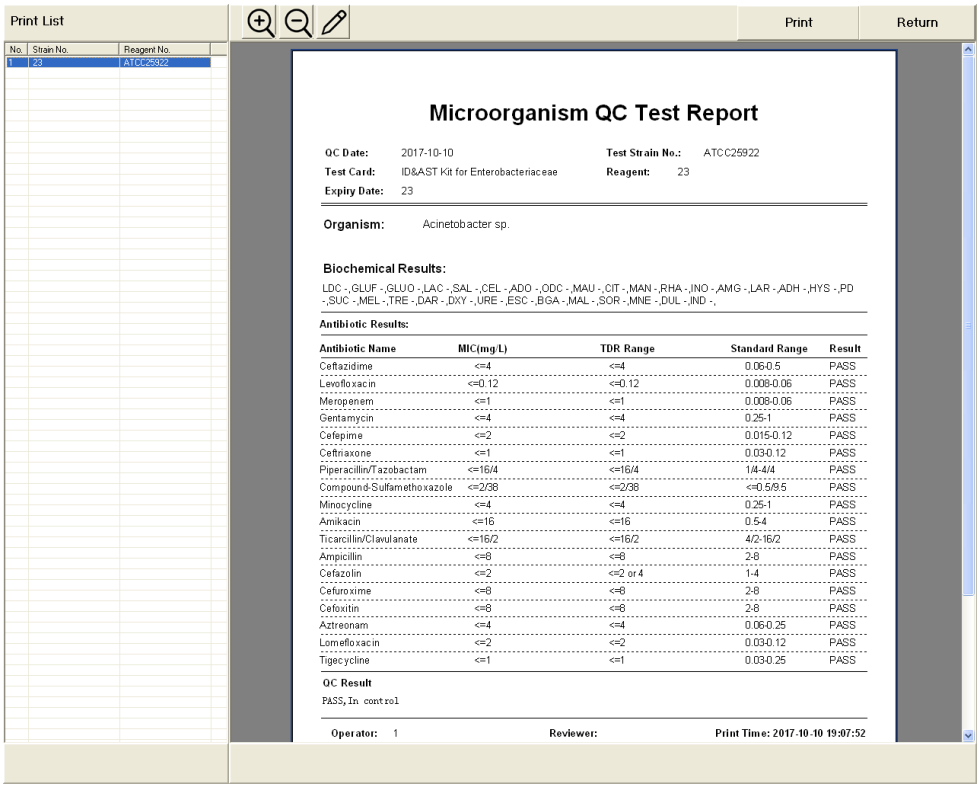


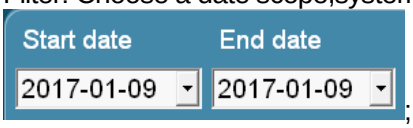
Fig 3-49

5. Click “ ”, it will zoom in the report; click “ ”, it will zoom out the report; click “ ”, it will go to the report template interface, you can adjust the template, then save it to take effect. Click “Print” button to print the report;
6. For the QC Results that you want to delete, check “√” to select them, then click “Delete”;
7. Click “Add”to do the next QC test, and then repeat steps 2-4;

3.5 Result

Result module contains the following functions:

1. Sort and filter the results. (See Fig 3-50):
   a) Filter: Choose a date scope,system will display the corresponding results.



;

   b) Sort: Click these function keys: Ascending, Descending, system will sort the results.;

Other keys: Collapse all, Expand All, Query;

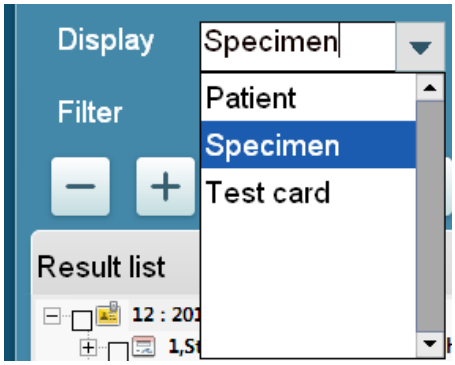


Fig 3–50

2. Display detailed results in result module: contain biochemical results, susceptibility results, original results, manual antibiotic results, Remark ;(See Fig 3-51)

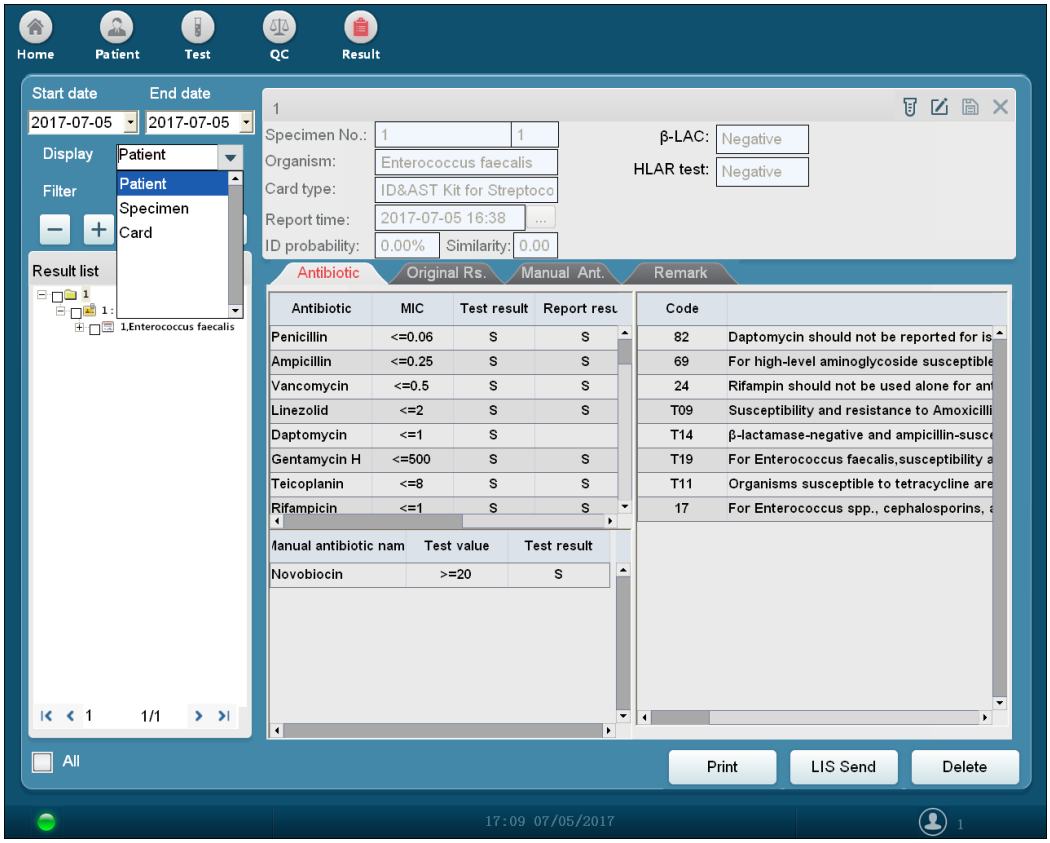


Fig 3–51

3. Review Results: Select a date scope, click the drop-down list of filter field, select “Waiting for review”, system will filter the un-reviewed results：
4. Approve Results: Select a date scope, click the drop-down list of filter field, select “Waiting for approval”, system will filter the unapproved results.
5. Print Results: Select a date scope, click the drop-down list of filter field, select “Waiting for print”, system will filter the unprinted results. User can print the results individually or in batches.
6. Send results to LIS system: Select a date scope, click the drop-down list of filter field, select “Waiting for LIS send”, system will filter the unsent results. User can send the results individually or in batches.
7. Remark: User can edit the remark content, it will be updated to test module.

Note：If the review, approval and LIS send function are not enabled, you can skip step 3, 4 and 6.

3.6 Export WHONET Data

WHONET module is intended to export the data from this software to the professional WHONET system to do the analysis and statistics.

Click “

” on the main screen to enter into the WHONET interface, see Fig 3-52:

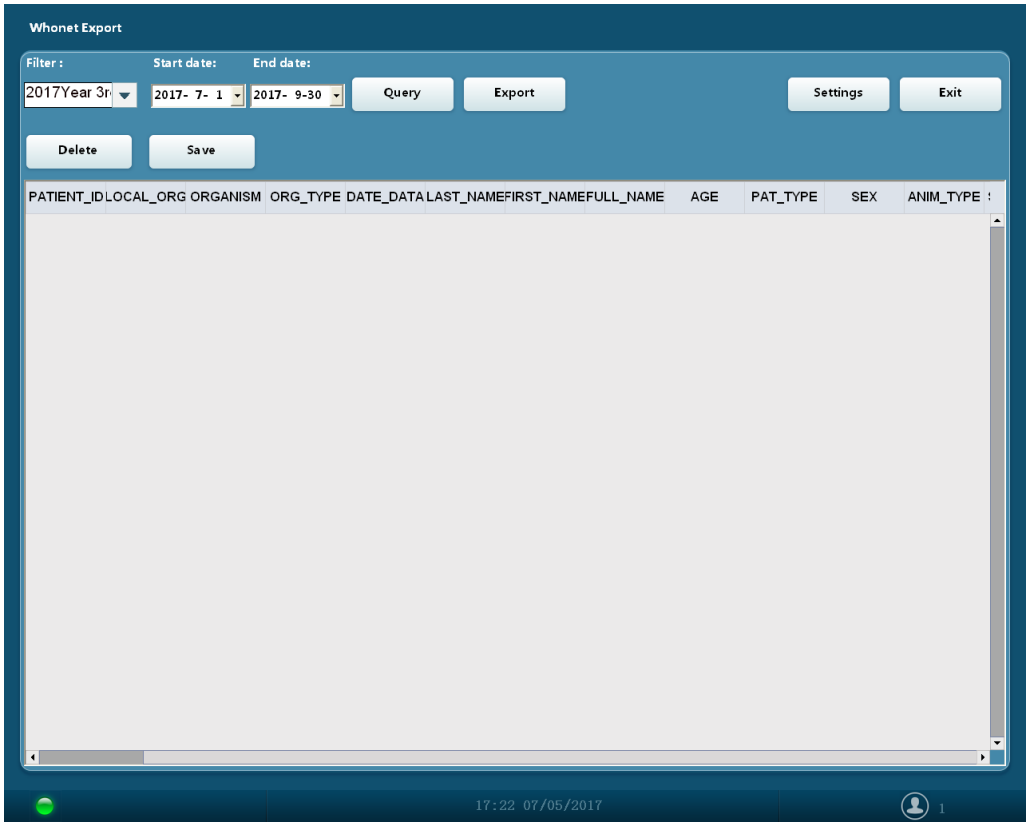


Fig 3–52

Procedures：

1. Click“

” to enter into the WHONET setting interface, user can set basic information, specimen info, department info and hospital info; the settings should be consistent with WHONET system.
2. Select the time period for the data that needs to be exported.
3. Click “Query”, system will list out the results.
4. System has the self-check function, if the required fields are blank, system will fill the blank fields with red color. User can double-click these fields to modify or add result, then click “Save”.
5. Click “Export”, it will pop out a dialog-box, select a path, input the file name, save it as dbf format.
6. Import the exported dbf file to WHONET system to do analysis and statistics.

3.7 Help

Help is intended to help you improve the knowledge of the software and the microorganism.

Procedure:

1. Click “Help” and select one option from the list, click “Open”.
2, System will open the corresponding help file.

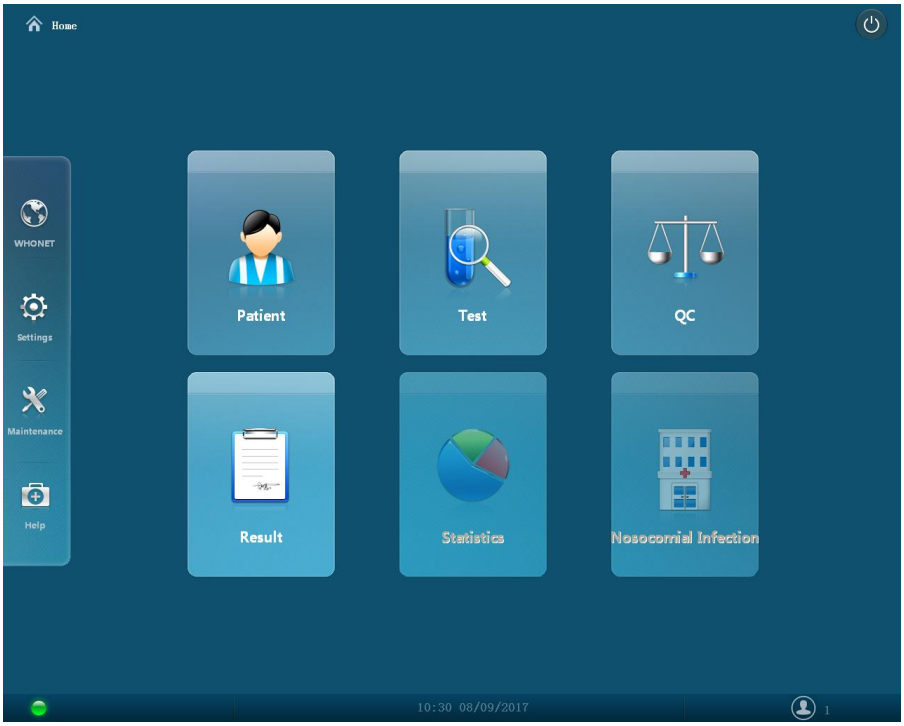


Fig 3–53

4 Test

4.1 Startup Test

1. Double click the icon of the TDR Microorganism System to enter the login window.
2. Enter the user name and password, click [Login].
3. Click “Patient", enter into the patient information screen.
4. Enter the patient information as specified in section 3.3.1.
5. Click the "test" icon; enter the bacterial identification and drug sensitivity test interface.

4.2 Test Procedure

1. Select an incubated test card and make sure the bottom is clean. .
2. Place the test card in the instrument and make sure the test card is pressed by the spring plate.
3. Add the patient info and go to the test module.
4. Choose the patient, and click “Auto” to start the test.

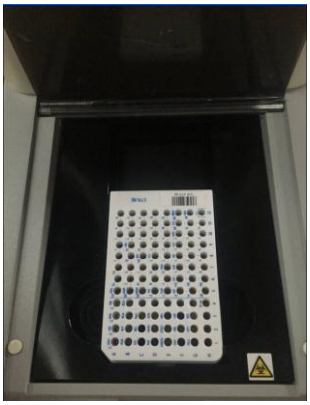


Fig 4.1

Note:

When the test card is placed on the rack, make sure it is even with the surface of the rack.

Before the test card is placed into the analyzer, make sure its bottom is clean and there is no solution or dirt on it.

When the rack is moving, do not put your hand into the analyzer.

Normal test procedure: when the test card is put in the rack, it is transferred inwards by the motor for several seconds to reach the test position. The test head measures the test card and acquires the test data. After analysis, the test results are obtained.

4.3 Test Precautions

1. Make sure the in-use test card matches the one set up in the software and is not expired; otherwise wrong test results may be produced.
2. The incubated test card should be tested timely on the analyzer otherwise the test result may be affected.
3. Make sure the bottom of the test card is clean, otherwise the remaining solution or dirt may affect the test result.
4. During the test, cover the transparent cover; otherwise the test result may be affected.
5. Clean the silkscreen plate on the rack regularly and make sure the silkscreen plate is clean.

5 Trouble shooting

Common Errors and Corrective Actions

1, The computer fails to work.

Cause 1: The power supply of the computer is disconnected.

Corrective Action: Check if the power supply of the computer is normal.

Cause 2: The operating system of the computer breaks down.

Corrective Action: Check if the operating system is normal. If it breaks down, it should be recovered by a professional computer maintenance engineer or reinstalled.

Cause 3: The computer contacts the virus.

Corrective Action: Use the anti-virus software to kill the virus.

Cause 4: Computer hardware errors.

Corrective Action: Please contact the computer's local after-sales service.

2, You cannot enter the system software

Cause 1: wrong user name and/or password.

Corrective Action 1: Enter correct user name and password.

Corrective Action 2: Please contact our company.

Cause 2: Property error of the application file.

Corrective Action: Please contact our company.

Cause 3: Folder name of the Microorganism Analysis System is wrong.

Corrective Action: Make sure the installation directory is correct.

Cause 4: System program files are incomplete.

Corrective Action: Please contact our company.

3 The system prompts Soft dog can’t be found.

Cause 1: soft dog is not properly connected.

Corrective Action: Connect again. If it is correctly connected, an icon can be seen on the lower side of the screen.

Cause 2: The driver program of the soft dog is lost.

Corrective Action: reinstall the driver program of the soft dog.

Cause 3: soft dog is burnt up.

Corrective Action: Please contact our company.

Cause 4: The USB ports of the main board are shut or damaged.

Corrective Action: Enter the CMOS program to enable the serial ports or contact our company.

4 The system can’t judge the data automatically.

Cause 1: The CCD signal cable and the collection card are not properly connected.

Corrective Action 1: Check the signal cable connection.

Corrective Action 2: Run One Touch Grabber program to see if it can acquire the images.

Cause 2: The PC and the collection card are not properly connected.

Corrective Action: Reconnect the USB cable.

Cause 3: When you click Auto Identification, the system prompts "Video initialization failed", "Serial port timeout".

Corrective Action 1: Exit the program and restart the system.

Corrective Action 2: Turn on the power of the identification device.

Cause 4: The driver program of the collection card is lost.

Corrective Action: Re-install the collect card driver or contact out company,

Cause 5: Collection card and camera error

Corrective Action: Please contact our company.

5 The judgment results varied widely

Cause 1: The judgment coordinates are not well calibrated or biased after the analyzer is moved.

Corrective Action: Recalibrate the coordinates.

Cause 2: The parameters of the analyzer are not set up correctly or have been modified.

Corrective Action: Please contact the engineer of our company to reset the parameters.

Cause 3: The acquired images are unclear and messy.

Corrective Action: Check if CCD and the signal cable of the collection card are properly connected.

6 The report can’t be printed

Cause 1: The printer and the computer are not connected.

Corrective Action: Check the following:

Ensure the power supply and the signal cable of the printer are correctly connected.

Ensure there is printer paper in the printer.

Ensure the printer paper is not jammed.

Cause 2: The driver program of the printers not installed.

Corrective Action: reinstall the driver program of the printer.

Cause 3: If multiple printers are being connected to the computer, the current printer is not the default one.

Corrective Action: Enter control panel - printer setup to set the current as the default one.

7 The report can’t be printed out normally.

Cause 1: If the printer model or the computer operating system has been changed, the driver program of the printer may not be compatible with the operating system.

Corrective Action: Reinstall the proper printer driver program for the printer and ensure it is compatible with the operating system.

Cause 2: Program error of the Microorganism Analysis System

Corrective Action: Please contact our company.

Cause 3: The print-out report is blank or unclear.

Corrective Action: change the cartridge or add toner.

8 Monitor error

Cause 1: The monitor fails to display.

Corrective action: ensure the power supply and the signal cable of the monitor are correctly connected.

Cause 2: Hardware error

Corrective Action 1: Please contact the computer's local after-sales service.

Corrective Action 2: Please contact our company.

6 Maintenance

6.1 Daily Maintenance and Precautions

- During test, please place the analyzer away from the direct sunlight otherwise the test precision may be affected.
- Use soft dry cloth to clean the device and make it clean once a week. If the panel of the device is dirty, please wipe it with clean water. Organic solvent such as gasoline, diluted paint, benzene compounds and so on are not allowed because they will deform and erode the analyzer, make the paint off and affect the performance and appearance.
- Do not use the water to clean the liquid crystal monitor instead use the soft clean dry cloth or tissue to wipe it.
- Make sure the rack is clean. Wipe the dust off gently with soft dry cloth once a week, because too much dust on the rack may affect the precision of the test.
- The device must only be operated by skilled/trained doctors, nurses or clinical professionals.

Note:

1. Due to the unique property of the test card, avoid using it at too high or too low environment temperature in case the result accuracy is affected. (Environment temperature 10-30℃).
2. You must use the test card that matches the analyzer so as to ensure the conformity and accuracy of the test results.
3. The disposal of the test card is subject to the local regulations of contamination and disposal.

BIOHAZARD:

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

Please read thoroughly this manual and the manual for the printer before using the system.

When the device is running, do not move or bump it and avoid spilling the bacteria solution into it.

Do not plug-in or unplug the power cord or signal connector without turning off the power supply. When shutting down the analyzer, return to the main screen.

Tools to be prepared by User:

Item	Applicable Maintenance
Soft dry cloth	Clean the rack and silkscreen plate.
Cotton swabs	Clean the rack and silkscreen plate.

6.2 Clean the Rack and Silkscreen Plate

Follow the steps below to clean the rack and silkscreen plate.

1. Turn off the power supply and open the transparent cover as shown in the following figure.

Fig 6.1

2. Use a soft cloth to wipe the internal groove and the surface of the silkscreen plate. Clean them gently. When you feel the resistance, do not push hard.

BIOHAZARD

Handle with caution and necessary protective measures.

6.3 Change the Fuse

Unplug the power cord. Use a flathead screwdriver to open the fuse box. Take out the fuse to see if it is burnt up. Replace it with a new one and close the fuse box.

Note:

Please use the fuse configured by our company or adopt the specification below: Diameter:5mm; length:20mm; T(Time lag) 250V/2AL.

Abbreviations:

ID, Identification;

No. , Number;

Info. , Information;

AST, Antimicrobial Susceptibility Testing;

QC, Quality Control;

ATCC, American Type Culture Collection;

Original Rs. , Original Results;

Manual Ant. , Manual Antibiotic;

Abbr. , Abbreviation;

S, Susceptible;

I, Intermediate;

R, Resistant.

Toxic and Harmful Substances or Elements and the Symbol of Their Content

Part Name	**Toxic and harmful substances or elements**					
	lead(Pb)	Mercury(Hg)	Cadmium(Cd)	Hexavalent Chromium(Cr((VI)	Polybrominated biphenyls(PBB)	Polybrominated diphenyl ethers (PBDE)
Bracket	○	○	○	○	○	○
Circuit board	×	○	○	○	○	○
Motor	○	○	○	○	○	○
Cable	○	○	○	○	○	○
Photometric system	○	○	○	○	○	○

○: shows that the content of the toxic and harmful substance contained in the part is below the required limit specified in SJ/T11363-2006 standards.

× : shows that the content of the toxic and harmful substance contained in the part is above the required limit specified in SJ/T11363-2006 standards.

Note: Applicable to TDR-300B PLUS.

046-004269-01

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